AAI_590_TommyPoole_CodeWithOutputs_clean

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1 Title: LLM RAG Distillation & Supervised Fine Tuning Optimization

1.1 Subtitle: Fine tuning open source nano models with larger reasoning models

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Attribution: The following AI tools were referenced for intial code drafting, as well as review, and troubleshooting through the actual development process: Claude Code March-April 2026. Perplexity.ai, accessed March -April 2026. Also Gemini, accessed March-April 2026. Also, Microsoft, this video was a key inspiration for this line of questioning related to RAG optimization approaches.

Microsoft Citation: Frame, A., More, O., & Juarez, S. (2025, September 8). There's no reason not to fine tune anymore [Video]. Microsoft AI Show. https://learn.microsoft.com/en-us/shows/ai-show/theres-no-reason-not-to-fine-tune-anymore

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1.2 Introduction, RAG

Retreival Augmented Generation (RAG) helps Large Language Models perform when applied to contexts with specific knowledge, such as enterprise business settings. RAG solutions produce more accurate information by gounding generated answers in a corpus of specific knowledge, vocuabluary, and even wriring style. One way this kind of application can be optimized to produce better answers is with supervised fine-tuning (SFT), where specific information is used to improve the base LLM response. This fine-tuning, coupled with RAG, can achieve even better results than RAG alone. One of the key questions is how much fine-tuning data is needed to improve results.

1.2.1 RAG Overview Graphic

Attribution: this graphic was made from text the author supplied to Claude, and refined through multiple iterations to organize in this diagram.

1.2.2 Project Outline

This project compares-fine tuning with different amounts of data across different models. The code is outlined as follows:

1. Import key libraries.
2. Download models from HuggingFace

- Medium Sized Reasoning Model (DeepSeek-R1-Distill-Qwen-14B)
- Nano Models for training (SmolLM3-3B, Nemotron 3 Nano-4B, Llama-3.2-1B-Instruct)
- Connect API to Frontier Model, Claude 4.6 for comparison, knowledge distillaiton, and evaluation scoring

3. Upload or ingest a corpus of data.
4. Generate text embeddings, store in vector store.
5. Use frontier model to create question answer pairs each based on the corpus, convert to JSONL format and combine in one dataset.
6. Create four datasets from those total questions of QA pairs for fine tuning of the following sizes: (40 held out as test), then 256 QA pairs, 64 QA pairs, and 8 QA pairs.
7. Train: Fine-tune all three nano models with the question answer pairs of varying sized datasets.
8. Generate Responses: Generate responses with trained models, and base models: Generate responses to the 40 held out questions with the three fine-tuned models, and all three non fine-tuned models base nano, with the same prompt and corpus for reitreival. Also generate answers with reasoning and frontier models against the same questions.
9. Evaluate: Use frontier model to evaluate each response against the ground truth for groundedness, relevance, and response completeness based on Microsoft documentation to determine scores on a 1-5 scale.
10. Visualize results.
11. Discuss results. The primary question for evaluation is whether or not the total number of questions used impacts the fine tuning results.

1.2.3 Project Outline Flow

Attribution: Created with the aide of Claude Sonnet 4.6

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2 1. Import Libraries

NOTE: If using Google CoLab, connect to GPU with High Ram. May require subscription.

[ ]:
```
# Install dependencies (run once)
!pip install transformers datasets huggingface_hub peft trl accelerate␣
↪bitsandbytes
!pip install sentence-transformers faiss-cpu
!pip install anthropic
!pip install jsonlines tqdm
!pip install pypdf
!pip install matplotlib seaborn
!pip install -q -U google-generativeai
!pip install pdfplumber pymupdf
```

```
!pip install --upgrade transformers accelerate bitsandbytes optimum
```

[2]:
```
# Standard Library
import os
import json
import re
from pathlib import Path
import sklearn
from sklearn.model_selection import train_test_split
import pypdf
import random, time
from peft import PeftModel
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
from math import pi
import pdfplumber
import fitz

# Third-party: data & numerics
import numpy as np
import pandas as pd
from tqdm import tqdm
import jsonlines
import gc
import hashlib
from typing import List, Dict
from sklearn.utils import resample

# PyTorch
import torch

# HuggingFace: models, tokenizers, datasets
from transformers import (
    AutoTokenizer,
    AutoModelForCausalLM,
    BitsAndBytesConfig,
    TrainingArguments,
    pipeline,
)
from datasets import Dataset, load_dataset
from huggingface_hub import login as hf_login

# HuggingFace: fine-tuning (PEFT / SFT)
from peft import LoraConfig, get_peft_model, TaskType
from trl import SFTTrainer, SFTConfig
```

```python
# Embeddings & vector store
from sentence_transformers import SentenceTransformer
import faiss

# Anthropic (Claude)
import anthropic

# Google CoLab
from google.colab import userdata

# Sanity check (print for check)
print(f"PyTorch version : {torch.__version__}")
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
if torch.cuda.is_available():
    print(f"GPU name   : {torch.cuda.get_device_name(0)}")


import warnings
warnings.filterwarnings(
    "ignore",
    message=r".*Both `max_new_tokens`.*and `max_length`.*"
)
```

```
PyTorch version : 2.10.0+cu128
Using device: cuda
GPU name   : NVIDIA RTX PRO 6000 Blackwell Server Edition
```

3 2. Model Setup & API Connections

[3]:
```python
# API Keys
HF_TOKEN      = userdata.get("HF_TOKEN")
ANTHROPIC_API_KEY = userdata.get("ANTHROPIC_KEY")

# --- Reasoning Models (Baseline Inference) ---
REASONING_MODELS = {
    "deepseek" : "deepseek-ai/DeepSeek-R1-Distill-Qwen-14B", # specifically␣
 ↪high rank on RAG
}

# --- Nano Models (fine-tune targets) ---
NANO_MODELS = {
    "smollm"   : "HuggingFaceTB/SmolLM3-3B",
    "nemotron" : "nvidia/Nemotron-Mini-4B-Instruct",
    "llamaNano" : "meta-llama/Llama-3.2-1B-Instruct",
}
```

```
# LoRA target modules - override per model if architecture differs
LORA_TARGETS = {
    "smollm"   :␣
 ↪["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"],
    "nemotron" :␣
 ↪["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"],
    "llamaNano" :␣
 ↪["q_proj","k_proj","v_proj","o_proj","gate_proj","up_proj","down_proj"],
}

# --- Dataset sizes for SFT comparison ---
DATASET_SIZES = [256, 64, 32, 8]
NUM_TEST      = 40 # held-out questions (never used in training)

# --- Data Directories ---
DATASET_DIR   = Path("data") # filepath containing source PDFs
DATASET_DIR.mkdir(parents=True, exist_ok=True)

# Derived subdirectories - all cascade from DATASET_DIR
QA_DIR      = DATASET_DIR / "qa_outputs" # ./data/qa_outputs/
SPLITS_DIR  = DATASET_DIR / "qa_splits" # ./data/qa_splits/
MODELS_DIR  = DATASET_DIR / "models" # ./data/models/
RESULTS_DIR = DATASET_DIR / "results" # ./data/results/
EVAL_DIR    = DATASET_DIR / "eval" # ./data/eval/

# Create all directories
for d in [QA_DIR, SPLITS_DIR, MODELS_DIR, RESULTS_DIR, EVAL_DIR]:
    d.mkdir(parents=True, exist_ok=True)

print(f"  Unified paths:")
print(f"  ROOT     : {DATASET_DIR}")
print(f"  QA       : {QA_DIR}")
print(f"  QA Splits: {SPLITS_DIR}")
print(f"  Models   : {MODELS_DIR}")
print(f"  Results  : {RESULTS_DIR}")
print(f"  Eval     : {EVAL_DIR}")

# --- API Connections ---
hf_login(token=HF_TOKEN)
print("\nHuggingFace login : OK")

# Claude
claude_client = anthropic.Anthropic(api_key=ANTHROPIC_API_KEY)
CLAUDE_MODEL  = "claude-sonnet-4-6"
_ping = claude_client.messages.create(
    model=CLAUDE_MODEL, max_tokens=16,
```

```
    messages=[{"role": "user", "content": "Reply OK"}],
)
print(f"Claude connected  : {_ping.content[0].text.strip()}")

# --- Shared 4-bit quantization config ---
bnb_4bit = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16,
    bnb_4bit_use_double_quant=True,
    llm_int8_enable_fp32_cpu_offload=True,
    llm_int8_threshold=6.0,
)

# Define Random Seed
SEED = 7 # adjusted from 24 to 7 to 13 per Proffessor Anna Marbut's␣
↪recommendation

print(f"\n RANDOM SEED = {SEED}")

print("\n  Config complete - models load on demand in later sections.")
```

```
 Unified paths:
  ROOT     : data
  QA       : data/qa_outputs
  QA Splits: data/qa_splits
  Models   : data/models
  Results  : data/results
  Eval     : data/eval

HuggingFace login : OK
Claude connected  : OK

 RANDOM SEED = 7

  Config complete - models load on demand in later sections.
```

4 3. Corpus Ingestion

Requires file(s) upload. Used key textbook for AAI Program as example.

Citation Needed:

[4]:
```
# Config
CHUNK_SIZE    = 384 # scaled down from 512 to reduce noise in chunks
CHUNK_OVERLAP = 96 # overlapping char number, to preserve context
```

```python
# Load chunks if already exist, don't duplicate below if able
def load_corpus_chunks(chunks_path: Path):
    if chunks_path.exists():
        with jsonlines.open(chunks_path) as f:
            chunks = list(f)
        print(f"  Loaded {len(chunks):,} saved chunks from {chunks_path}")
        return chunks
    return None

# --- PDF Text Extraction ---

# Use PDF Plumber for more robust text extraction, fail to fitz if pdfplumber␣
↪fails
# *Returns* concatonated text with page markers
def extract_text_from_pdf(pdf_path: Path) -> str:
    text_parts = []
    try:
        with pdfplumber.open(str(pdf_path)) as pdf:
            num_pages = len(pdf.pages)
            print(f"  Extracting {pdf_path.name} ({num_pages:,} pages)...")
            for i, page in enumerate(pdf.pages):
                page_text = page.extract_text() or ""
                if page_text.strip():
                    text_parts.append(f"--- Page {i+1} ---\n{page_text}")
                page.flush_cache()
                if (i + 1) % 100 == 0:
                    print(f"    Page {i+1}/{num_pages}")
            full_text = "\n\n".join(text_parts)
            print(f"  {pdf_path.name}: {num_pages:,} pages → {len(full_text):,}␣
↪chars")
            return full_text
    except Exception as e:
        print(f"  pdfplumber failed: {e}")

    # Secondary fallback to fitz (PyMuPDF)
    try:
        doc = fitz.open(str(pdf_path))
        num_pages = len(doc)
        print(f"  Falling back to PyMuPDF for {pdf_path.name} ({num_pages:,}␣
↪pages)...")
        for i, page in enumerate(doc):
            page_text = page.get_text()
            if page_text.strip():
                text_parts.append(f"--- Page {i+1} ---\n{page_text}")
            if (i + 1) % 100 == 0:
                print(f"    Page {i+1}/{num_pages}")
        doc.close()
```

```
        full_text = "\n\n".join(text_parts)
        print(f"  {pdf_path.name}: {num_pages:,} pages → {len(full_text):,}␣
↪chars")
        return full_text
    except Exception as e:
        print(f"  PyMuPDF failed: {e}")
        return f"[EXTRACTION FAILED: {pdf_path.name}]"


# --- Text Chunking ---

# Split extracted text into semantically cohesive chunks
# Handles paragraph boundaries, overlaps, and filters small chunks
def chunk_text(text: str, chunk_size: int = CHUNK_SIZE, overlap: int =␣
↪CHUNK_OVERLAP) -> list[str]:
    paragraphs = re.split(r'\n\s*\n|\n#{1,3}\s', text)
    chunks = []

    # Split text using blank lines or headers as separators
    for para in paragraphs:
        para = para.strip()
        if not para:
            continue

        # If paragraph fits chunk size, keep as-is
        if len(para) <= chunk_size:
            chunks.append(para)
        else:
            # Otherwise, split with overlaping windoes to maintain semantic␣
↪continuity
            start = 0
            while start < len(para):
                chunk = para[start:start + chunk_size].strip()
                if chunk:
                    chunks.append(chunk)
                start += chunk_size - overlap

    # Filter out very short fragments (they are likely noise)
    return [c for c in chunks if len(c) > 50]


# --- Corpus Assembly ---
chunks_file = DATASET_DIR / "corpus_chunks.jsonl"
corpus_chunks = load_corpus_chunks(chunks_file)

if corpus_chunks is None:
```

```
    # Locate PDF Files
    pdf_files = sorted(DATASET_DIR.glob("*.pdf"))
    assert pdf_files, f"No PDFs found in {DATASET_DIR.resolve()}"

    corpus_chunks = []

    # Extract and chunk each PDF
    for pdf_path in pdf_files:
        raw_text = extract_text_from_pdf(pdf_path)
        chunks = chunk_text(raw_text)
        for i, chunk in enumerate(chunks):
            corpus_chunks.append({
                "source": pdf_path.name,
                "chunk_id": i,
                "text": chunk
            })
        print(f"  → {len(chunks):,} chunks from {pdf_path.name}")

    # Save all generated chunks to disk
    with jsonlines.open(chunks_file, "w") as f:
        f.write_all(corpus_chunks)
    print(f"\n\n\n  Saved {len(corpus_chunks):,} chunks to {chunks_file}")


# --- Corpus Statistics ---

# Count tokens for overview of chunk size distribution
_count_tok = AutoTokenizer.from_pretrained(list(NANO_MODELS.values())[0],␣
 ↪token=HF_TOKEN)
total_tokens = 0
total_chars = 0
for chunk in corpus_chunks:
    total_chars += len(chunk["text"])
    total_tokens += len(_count_tok(chunk["text"],␣
 ↪add_special_tokens=False)["input_ids"])
del _count_tok

print(f"  Avg chunk tokens: {total_tokens / len(corpus_chunks):.0f}")

# Preview a few randomly sampled chunks for sanity check
samples = random.sample(corpus_chunks, min(5, len(corpus_chunks)))
print("\n\n\n=== 5 RANDOM CHUNK SAMPLES ===")
for i, chunk in enumerate(samples, 1):
    print(f"\n{i}. {chunk['source']} (chunk {chunk['chunk_id']})")
    print(f"   {repr(chunk['text'][:200])}...")

# Display overall corpus statistics
```

```python
print(f"\n=== CORPUS SUMMARY ===")
print(f"  PDFs loaded   : {len(set(c['source'] for c in corpus_chunks))}")
print(f"  Total chunks  : {len(corpus_chunks):,}")
print(f"  Total chars   : {total_chars:,}")
print(f"  Total tokens  : {total_tokens:,}")
```

```
 Loaded 11,773 saved chunks from data/corpus_chunks.jsonl
  Avg chunk tokens: 102



=== 5 RANDOM CHUNK SAMPLES ===

1. Artificial Intelligence - A Modern Approach (3rd Edition).pdf (chunk 2034)
   'rofthe2008ComputerOlympiad.\nGoisadeterministic
game,butthelargebranchingfactormakesitchalleging. Thekey\nissuesandearlyliteratu
reincomputerGoaresummarizedbyBouzyandCazenave(2001)and\nMu¨ller (2002). Up '…

2. Artificial Intelligence - A Modern Approach (3rd Edition).pdf (chunk 3889)
   '--- Page 408 ---\nSection10.4. OtherClassicalPlanningApproaches 389\nGold
PIT\nPIT\nPIT\nGold PIT\nPIT\nPIT\nGold PIT\nPIT\nPIT\nResult(Result(S,
Forward),\n0\nTurn(Right))\nTurn(Right)\nResult(S, Forward)\n0\nForward'…

3. Artificial Intelligence - A Modern Approach (3rd Edition).pdf (chunk 5248)
   'ATION\nVariableelimination worksbyevaluating expressions such
asEquation(14.4)inright-to-left\norder(thatis,bottomupinFigure14.8).
Intermediateresultsarestored,andsummationsover\neachvariable aredoneonly'…

4. Artificial Intelligence - A Modern Approach (3rd Edition).pdf (chunk 4675)
   'fers that\nhavebeenextracted. Forexample,considerthesethreeoffers:\nA
:1.4GHzCPU,2GBRAM,250GBdisk,$299.\nB : 1.2GHzCPU,4GBRAM,350GBdisk,$500.\nC :
1.2GHzCPU,2GBRAM,250GBdisk,$399.\nC is dominated by A; tha'…

5. Artificial Intelligence - A Modern Approach (3rd Edition).pdf (chunk 5527)
   'VINCE (Kim, 1983). Early\napplicationsinmedicineincludedtheMUNINsystemfordia
gnosingneuromusculardisorders\n(Andersen etal.,1989) andthe PATHFINDER system
forpathology (Heckerman, 1991). The\nCPCS system '…

=== CORPUS SUMMARY ===
  PDFs loaded   : 1
  Total chunks  : 11,773
  Total chars   : 4,301,167
  Total tokens  : 1,195,207
```

5 4. Embeddings & Vector Store

Retrieval setup, turning pre-chunked corpus into a searchable knowledge base using sentence embeddings and FAISS, Facebook AI Similarity Search for effecient similarity searching.

[5]:
```
# Config
EMBED_MODEL_ID = "all-MiniLM-L6-v2"   # fast, reliable; can swap for larger␣
↪model if needed, creates 384-dimensional embeddigns
TOP_K          = 5                     # default chunks to retrieve per query

# --- Model Loading ---
# Load embedding model, SentenceTransformer, maps text chunks or queries into␣
↪dense vector embeddings
print(f"Loading embedding model: {EMBED_MODEL_ID} ...")
embed_model = SentenceTransformer(EMBED_MODEL_ID)
# Obtain model's embedding dimensionality to configure FAISS later
EMBED_DIM   = embed_model.get_sentence_embedding_dimension()
print(f"Embedding dimension: {EMBED_DIM}")

# --- Encoding Corpus ---

# Extract raw text of all chunks into a simple list
print(f"Encoding {len(corpus_chunks):,} chunks ...")
chunk_texts = [c["text"] for c in corpus_chunks]

# Compute embeddings for each chunk in batches of 64
# normalize embeddings=True ensures embeddings have unit length for dot product␣
↪later
chunk_embeddings = embed_model.encode(
    chunk_texts,
    batch_size=64,
    show_progress_bar=True,
    normalize_embeddings=True, # required for cosine similarity via inner␣
↪product
    convert_to_numpy=True,  # FAISS expects NumPy arrays
)

# --- FAISS Index Building ---

# Build FAISS index (cosine similarity via inner product on normalized vecs)
faiss_index = faiss.IndexFlatIP(EMBED_DIM)

# Store all embeddings into the index for retrieval
faiss_index.add(chunk_embeddings)
print(f"FAISS index built: {faiss_index.ntotal:,} vectors")

# --- Retrieval Function ---
```

```python
# Retrieve top-k most semantically relavant chunks for a given text query
# *Return* Top-k chunks with original metadat and similarity score
def retrieve(query: str, k: int = TOP_K) -> list[dict]:

    # Encode query into same embedding sapce
    query_vec = embed_model.encode(
        [query],
        normalize_embeddings=True,
        convert_to_numpy=True,
    )

    # Perform similarity search: returns (scores, indices)
    scores, indices = faiss_index.search(query_vec, k)
    results = []

    # Gather metadata
    for score, idx in zip(scores[0], indices[0]):
        results.append({**corpus_chunks[idx], "score": float(score)})
    return results

# --- Smoke Test ---
# Simple question to see how well it worked
test_query   = "What is artificial intelligence?"
test_results = retrieve(test_query)

print ("\n\n\n=== SMOKE TEST ===")
print(f"\nSmoke test - query: '{test_query}'")
for r in test_results:
    print(f"  [{r['score']:.3f}] {r['source']} chunk {r['chunk_id']}:␣
↪{r['text'][:80]}...")
```

```
Loading embedding model: all-MiniLM-L6-v2 …

Loading weights:   0%|          | 0/103 [00:00<?, ?it/s]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Embedding dimension: 384
Encoding 11,773 chunks …

Batches:   0%|          | 0/184 [00:00<?, ?it/s]

FAISS index built: 11,773 vectors
```

```
=== SMOKE TEST ===

Smoke test - query: 'What is artificial intelligence?'
  [0.686] Artificial Intelligence - A Modern Approach (3rd Edition).pdf chunk
127: so important
INTELLIGENCE
tous. Forthousands ofyears, wehavetried tounderstand h…
  [0.666] Artificial Intelligence - A Modern Approach (3rd Edition).pdf chunk
426: r,inSimon'swords,"machinesthatthink,thatlearn
andthatcreate." Theycalltheeffort …
  [0.653] Artificial Intelligence - A Modern Approach (3rd Edition).pdf chunk
452: ords(Brants etal.,2007). Noneofthecomputerscientists
ontheteamspeakArabic,butthe…
  [0.630] Artificial Intelligence - A Modern Approach (3rd Edition).pdf chunk
321: es. The second answer is methodology. AI is the only one
ofthesefieldsthatisclea…
  [0.622] Artificial Intelligence - A Modern Approach (3rd Edition).pdf chunk
126: --- Page 20 ---
1
INTRODUCTION
In which we try to explain why we consider artifi…
```

6 5. QA Pair Generation & Organization

[ ]:
```python
# Output File Path
combined_file = QA_DIR / "qa_claude_only.jsonl"

# Hyperparams
TARGET_QA = 300
SAMPLE_CHUNK_SIZE = 200  # how many chunks to sample per batch
BATCH_SIZE = 8           # how many calls per loop
MAX_NEW_TOKENS = 192
TEMPERATURE = 0.1
DEBUG_RAW = False
```

[ ]:
```python
# Deterministic Random Number generator
def get_rng(seed: int, offset: int = 0) -> random.Random:
    """Stable per-run, deterministic RNG."""
    return random.Random(seed + offset)

# Extract a structured QA JSON object
def extract_json(text: str) -> dict | None:
    patterns = [
        r'\{.*?"question".*?"answer".*?\}',
```

```
        r'\{[^{}]*\}',
    ]
    for pattern in patterns:
        for m in re.findall(pattern, text, re.DOTALL):
            try:
                parsed = json.loads(m)
                if (
                    isinstance(parsed, dict)
                    and isinstance(parsed.get("question"), str)
                    and isinstance(parsed.get("answer"), str)
                    and parsed["question"].strip()
                    and parsed["answer"].strip()
                ):
                    return {
                        "question": parsed["question"].strip(),
                        "answer": parsed["answer"].strip(),
                    }
            except json.JSONDecodeError:
                continue
    # If no pattern yields valid QA, return None.
    return None
```

[ ]:
```
# Generate one QA pair per passage using claude
def generate_qa_batch_claude(
    batch: List[Dict],
) -> List[Dict]:
    QA_PROMPT = """\
Generate EXACTLY ONE question-answer pair as valid JSON.
Return exactly:
{{"question": "Your question?", "answer": "Your answer from passage only"}}

Use only the passage below.

PASSAGE:
{passage}
"""

    new_rows = []

    for p in batch:
        # Extract and clean the passage text, strip leading trailing whitespace
        passage = p["text"].strip()
        # Format the QA prompt with the current passage
        prompt = QA_PROMPT.format(passage=p["text"].strip())

        if DEBUG_RAW:
            print(f"[DEBUG CLAUDE PROMPT] {repr(prompt[:300])}")
```

```
        # Call the Claude API with:
        # - model name defined in CLAUDE_MODEL
        # - token limit MAX_NEW_TOKENS
        # - fixed temperature for consistent generation
        # - a single user message containing the QA prompt
        msg = claude_client.messages.create(
            model=CLAUDE_MODEL,
            max_tokens=MAX_NEW_TOKENS,
            temperature=TEMPERATURE,
            messages=[{"role": "user", "content": prompt}],
        )

        raw = "".join(
            block.text for block in msg.content
            if getattr(block, "type", None) == "text"
        ).strip()

        # Print full prompt for debugging, truncated to 300 chars
        if DEBUG_RAW:
            print(f"[DEBUG CLAUDE RAW] {repr(raw[:300])}")

        qa = extract_json(raw)

        if qa:
            new_rows.append({
                "id": len(new_rows),
                "question": qa["question"],
                "answer": qa["answer"],
                "source": p.get("source", "unknown"),
                "chunk_id": p["chunk_id"],
                "source_text": p["text"],
                "generator": "claude",
            })

        # Gentle rate-limiting
        time.sleep(0.15)

    return new_rows
```

[ ]:
```
# Check existing file, then load passages
PASSAGES_JSONL = Path("data/corpus_chunks.jsonl")

def load_jsonl(path: Path) -> List[Dict]:
    rows = []
    with open(path, "r", encoding="utf-8") as f:
```

```
        for line in f:
            line = line.strip()
            if line:
                rows.append(json.loads(line))
    return rows

if not PASSAGES_JSONL.exists():
    raise FileNotFoundError(
        f"Passages file not found at {PASSAGES_JSONL.resolve()}\n"
        f"Current dir: {os.getcwd()}\n"
        f"Files present: {os.listdir('.')}"
    )

passages = load_jsonl(PASSAGES_JSONL)
```

[ ]:
```
# Function to run generation until target is hit
def generate_until_target_claude(
    passages: List[Dict],
    target: int = 300,
    batch_size: int = 8,
) -> List[Dict]:
    rng = get_rng(SEED, offset=100)  # stable across runs
    pool = list(passages)           # copy
    rng.shuffle(pool)               # reproducible shuffle

    qa_pairs = []
    cursor = 0

    with tqdm(total=target, desc="Claude QA gen") as pbar:
        while len(qa_pairs) < target and cursor < len(pool):
            # Sample a batch from pool
            n = min(batch_size, target - len(qa_pairs))
            batch = pool[cursor:cursor + n]
            cursor += n

            rows = generate_qa_batch_claude(batch)
            qa_pairs.extend(rows)

            pbar.n = min(len(qa_pairs), target)
            pbar.refresh()

            if len(qa_pairs) >= target:
                break

    print(f"Generated {len(qa_pairs)} QA pairs (target: {target})")
    return qa_pairs[:target]
```

[ ]:
```
# --- Dry Run ---
# Use to ensure functioning as expected before larger token run
OUTPUT_FILE = QA_DIR / "qa_claude_300.jsonl"  # keep the same path


if OUTPUT_FILE.exists():
    print(f"Loading cached QA from {OUTPUT_FILE}")
    all_qa = load_jsonl(OUTPUT_FILE)
else:
    print("DRY-RUN: Generating 5 QA pairs with Claude (no HF models)...")
    all_qa = generate_until_target_claude(passages, target=5,␣
 ↪batch_size=BATCH_SIZE)

    # Append global IDs if you want
    for i, row in enumerate(all_qa):
        row["global_id"] = i

    print(f"DRY-RUN output: {len(all_qa):,} QA pairs")
    print("Preview:")
    for row in all_qa:
        print(f"  Q: {row['question'][:80]}...")
        print(f"  A: {row['answer'][:80]}...")

    # Optional: don't write to disk yet
    # print("Skipping disk write; rerun with target=300 when ready.")
```

[ ]:
```
# --- EXECUTION: Generate 300 QA from Claude ---

OUTPUT_FILE = QA_DIR / "qa_claude_300.jsonl"

if OUTPUT_FILE.exists():
    print(f"Loading cached QA from {OUTPUT_FILE}")
    all_qa = load_jsonl(OUTPUT_FILE)
else:
    print("Generating 300 QA pairs with Claude (no HF models)...")
    all_qa = generate_until_target_claude(passages, target=300,␣
 ↪batch_size=BATCH_SIZE)

    # Append global IDs
    for i, row in enumerate(all_qa):
        row["global_id"] = i

    print(f"Saving {len(all_qa):,} QA pairs to {OUTPUT_FILE}")
    with open(OUTPUT_FILE, "w", encoding="utf-8") as f:
        for row in all_qa:
            f.write(json.dumps(row, ensure_ascii=False) + "\n")
```

7 Create QA Subsets

Question/Answer Subsets

- Create Four QA subsets of different sizes (256, 64, 8), and test set of 40

[7]:
```
# Config Dataset Splitting
# Creates four sized training splits (256 / 64 / 8) and the 40-question
# held-out test set. All written to disk - re-run without regenerating QA.

# Load combined QA from disk if not already in memory
if "all_qa_pairs" not in dir() or not all_qa_pairs:
    with jsonlines.open(QA_DIR / "qa_claude_300.jsonl") as reader:
        all_qa_pairs = list(reader)
    print(f"Loaded {len(all_qa_pairs)} QA pairs from disk.")
    # Use Claude-only QA instead of mixed HF+Claude
    with jsonlines.open(QA_DIR / "qa_claude_300.jsonl") as reader:
        all_qa_pairs = list(reader)
    print(f"Loaded {len(all_qa_pairs)} Claude-only QA pairs from disk.")

random.seed(SEED)
shuffled   = all_qa_pairs.copy()
random.shuffle(shuffled)

# Reserve NUM_TEST for held-out test set
test_pairs = shuffled[:NUM_TEST]
train_pool = shuffled[NUM_TEST:]  # 300 available from generation

# Save test set
test_file = SPLITS_DIR / "test_40.jsonl"
with jsonlines.open(test_file, mode="w") as writer:
    writer.write_all(test_pairs)
print(f"Test set  ({len(test_pairs):>3} questions) → {test_file}")

# Create and save each training split
train_splits = {}
for size in DATASET_SIZES:
    n      = min(size, len(train_pool))
    subset = train_pool[:n]
    train_splits[size] = subset
    split_file = SPLITS_DIR / f"train_{size}.jsonl"
    with jsonlines.open(split_file, mode="w") as writer:
        writer.write_all(subset)
    print(f"  Train [{size:>3}q] → {split_file}  ({len(subset)} pairs)")

print(f"\n  All splits saved to {SPLITS_DIR}")
print(f"   Training sizes: {[len(v) for v in train_splits.values()]} | Test:␣
 ↪{len(test_pairs)}")
```

```
Loaded 300 QA pairs from disk.
Loaded 300 Claude-only QA pairs from disk.
Test set  ( 40 questions) → data/qa_splits/test_40.jsonl
  Train [256q] → data/qa_splits/train_256.jsonl  (256 pairs)
  Train [ 64q] → data/qa_splits/train_64.jsonl  (64 pairs)
  Train [ 32q] → data/qa_splits/train_32.jsonl  (32 pairs)
  Train [  8q] → data/qa_splits/train_8.jsonl  (8 pairs)

 All splits saved to data/qa_splits
  Training sizes: [256, 64, 32, 8] | Test: 40
```

8 7. Training: Supervised Fine-Tuning (LoRA + SFT)

LoRA, Low-Rank Adaptation, is a technique where all pre-trained weights are frozen and a subset of trainable paramters are injected into the architecture and then trained on specific example sets.

- Fine-tunes every nano model on every dataset size: 3 models × 3 sizes = 9 runs.
- Completed runs are skipped unless FORCE_RETRAIN=True.
- Each fine-tuned model saved to: models/{nano_key}*ft*{size}q/

[8]:
```
# Config

FORCE_RETRAIN = False

# --- LoRA hyperparameters ---
LORA_R        = 16
LORA_ALPHA    = 32
LORA_DROPOUT  = 0.15 # increased from .5 to address overfitting on QA artifacts␣
 ↪for larger dataset

# --- Training hyperparameters ---
NUM_EPOCHS    = 3
BATCH_SIZE    = 4
GRAD_ACCUM    = 4        # effective batch size = 16
LEARNING_RATE = 2e-4
MAX_SEQ_LEN   = 512
TRAIN_SPLIT   = 0.9

# --- Load splits from disk ---
QA_DIR     = DATASET_DIR / "qa_outputs"
SPLITS_DIR = DATASET_DIR / "qa_splits"


# load immutable 40 questoin test set
with jsonlines.open(SPLITS_DIR / "test_40.jsonl") as r:
    test_pairs = list(r)

# Load each training split into memory
```

```python
train_splits = {}
for size in DATASET_SIZES:
    with jsonlines.open(SPLITS_DIR / f"train_{size}.jsonl") as r:
        train_splits[size] = list(r)

# Print dataset overview
print(f"Test set  : {len(test_pairs)} questions")
for size, pairs in train_splits.items():
    print(f"  Train [{size:>3}q]: {len(pairs)} pairs")

# --- Chat Formatter ---
# Convert single QA pair chat formatted training text
def format_chat(pair: dict, tokenizer) -> str:
    messages = [
        {"role": "user",      "content": pair["question"]},
        {"role": "assistant", "content": pair["answer"]},
    ]
    return tokenizer.apply_chat_template(
        messages, tokenize=False, add_generation_prompt=False,
    )

# --- Core SFT Function ---
# Fine tune one nano model on on dataset size
# Flow: load, format, LoRA, train, save, cleanup
def run_sft(nano_key: str, nano_id: str, size: int, pairs: list[dict]) -> Path:

    # Deterministic output dir
    out_dir = MODELS_DIR / f"{nano_key}_ft_{size}q"

    # Cache check, if file exists
    if out_dir.exists() and not FORCE_RETRAIN:
        print(f"Skipping {nano_key} × {size}q - checkpoint exists at {out_dir}")
        return out_dir

    print(f"\n{' '*60}")
    print(f"  Fine-tuning : {nano_key} | dataset size : {size}q")
    print(f"  Model ID    : {nano_id}")
    print(f"{' '*60}")

    # Load base model + tokenizer
    tok = AutoTokenizer.from_pretrained(nano_id, token=HF_TOKEN)
    tok.pad_token = tok.eos_token # required for batching

    base_model = AutoModelForCausalLM.from_pretrained(
        nano_id,
        torch_dtype=torch.bfloat16, # more effecient
        device_map="auto", # GPU/CPU auto-placement
```

```
    token=HF_TOKEN,
)

# --- Format & Split ---
texts = [format_chat(p, tok) for p in pairs]
train_texts, val_texts = train_test_split(
    texts, train_size=TRAIN_SPLIT, random_state=SEED
)
train_ds = Dataset.from_dict({"text": train_texts})
val_ds   = Dataset.from_dict({"text": val_texts})

# --- LoRA Config ---
lora_cfg = LoraConfig(
    r=LORA_R, # Rank
    lora_alpha=LORA_ALPHA, # Scaling
    lora_dropout=LORA_DROPOUT, # Regularization
    target_modules=LORA_TARGETS[nano_key], # Model-specific
    task_type=TaskType.CAUSAL_LM, bias="none", # Don't train bias terms
)
ft_model = get_peft_model(base_model, lora_cfg)
ft_model.print_trainable_parameters()

# --- Training Config ---
sft_cfg = SFTConfig(
    output_dir=str(out_dir),
    logging_dir=str(MODELS_DIR / "logs"),
    num_train_epochs=NUM_EPOCHS,
    per_device_train_batch_size=BATCH_SIZE,
    per_device_eval_batch_size=BATCH_SIZE,
    gradient_accumulation_steps=GRAD_ACCUM, # effective batch size
    learning_rate=LEARNING_RATE,
    lr_scheduler_type="cosine",
    warmup_ratio=0.05,
    bf16=True, # precision
    max_length=MAX_SEQ_LEN, # truncate long sequences
    dataset_text_field="text", # Uses formatted texts
    eval_strategy="epoch",
    save_strategy="epoch",
    load_best_model_at_end=True,
    logging_steps=10,
    report_to="none",
)

# --- Train ---
trainer = SFTTrainer(
    model=ft_model,
    args=sft_cfg,
```

```
        train_dataset=train_ds,
        eval_dataset=val_ds,
        processing_class=tok, # handles tokenization/padding
    )
    result = trainer.train()
    print(f"  Loss: {result.training_loss:.4f}  |  Runtime: {result.
 ↪metrics['train_runtime']:.0f}s")

    # --- Save ---
    out_dir.mkdir(parents=True, exist_ok=True)
    trainer.save_model(str(out_dir))
    tok.save_pretrained(str(out_dir))
    print(f"  Saved → {out_dir}")

    # --- GPU Cleanup ---
    # Unload before next run
    del ft_model, base_model, tok, trainer
    torch.cuda.empty_cache()
    return out_dir


# --- Main Training Loop: # nano models × # dataset sizes ---
total_runs   = len(NANO_MODELS) * len(DATASET_SIZES)
run_num      = 0
ft_model_dirs = {}  # {nano_key: {size: Path}}

print(f"\n\n\nStarting {total_runs} fine-tuning runs...")
print(f"   Models: {list(NANO_MODELS.keys())}")
print(f"   Sizes : {DATASET_SIZES}")

for nano_key, nano_id in NANO_MODELS.items():
    ft_model_dirs[nano_key] = {}
    for size in DATASET_SIZES:
        run_num += 1
        print(f"\n[Run {run_num}/{total_runs}]  {nano_key}  ×  {size}q")
        ft_dir = run_sft(nano_key, nano_id, size, train_splits[size])
        ft_model_dirs[nano_key][size] = ft_dir

print(f"\n  SFT complete - {total_runs} runs ({len(NANO_MODELS)} models ×␣
 ↪{len(DATASET_SIZES)} sizes)")
for nano_key, sizes in ft_model_dirs.items():
    for size, path in sizes.items():
        print(f"  {nano_key}_ft_{size}q → {path}")
```

```
Test set  : 40 questions
  Train [256q]: 256 pairs
  Train [ 64q]: 64 pairs
```

```
  Train [ 32q]: 32 pairs
  Train [  8q]: 8 pairs



Starting 12 fine-tuning runs…
   Models: ['smollm', 'nemotron', 'llamaNano']
   Sizes : [256, 64, 32, 8]

[Run 1/12]  smollm  ×  256q


  Fine-tuning : smollm | dataset size : 256q
  Model ID    : HuggingFaceTB/SmolLM3-3B


`torch_dtype` is deprecated! Use `dtype` instead!

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading (incomplete total…): 0.00B [00:00, ?B/s]

Fetching 2 files:   0%|          | 0/2 [00:00<?, ?it/s]

Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]

generation_config.json:   0%|          | 0.00/182 [00:00<?, ?B/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 30,228,480 || all params: 3,105,327,104 || trainable%: 0.9734

Adding EOS to train dataset:   0%|          | 0/230 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/230 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/26 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/26 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'bos_token_id': None, 'pad_token_id': 128012}.

<IPython.core.display.HTML object>

  Loss: 0.3748  |  Runtime: 41s
  Saved → data/models/smollm_ft_256q

[Run 2/12]  smollm  ×  64q
```

```
  Fine-tuning : smollm | dataset size : 64q
  Model ID    : HuggingFaceTB/SmolLM3-3B

Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 30,228,480 || all params: 3,105,327,104 || trainable%: 0.9734

Adding EOS to train dataset:   0%|          | 0/57 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/57 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'bos_token_id': None, 'pad_token_id': 128012}.

<IPython.core.display.HTML object>

  Loss: 0.5523  |  Runtime: 11s
  Saved → data/models/smollm_ft_64q

[Run 3/12]  smollm  ×  32q


  Fine-tuning : smollm | dataset size : 32q
  Model ID    : HuggingFaceTB/SmolLM3-3B

Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 30,228,480 || all params: 3,105,327,104 || trainable%: 0.9734

Adding EOS to train dataset:   0%|          | 0/28 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/28 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/4 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/4 [00:00<?, ? examples/s]
```

```
The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'bos_token_id': None, 'pad_token_id': 128012}.

<IPython.core.display.HTML object>

  Loss: 0.7381  |  Runtime: 6s
  Saved → data/models/smollm_ft_32q

[Run 4/12]  smollm  ×  8q


  Fine-tuning : smollm | dataset size : 8q
  Model ID    : HuggingFaceTB/SmolLM3-3B


Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 30,228,480 || all params: 3,105,327,104 || trainable%: 0.9734

Adding EOS to train dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/1 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/1 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'bos_token_id': None, 'pad_token_id': 128012}.

<IPython.core.display.HTML object>

  Loss: 0.8856  |  Runtime: 3s
  Saved → data/models/smollm_ft_8q

[Run 5/12]  nemotron  ×  256q


  Fine-tuning : nemotron | dataset size : 256q
  Model ID    : nvidia/Nemotron-Mini-4B-Instruct


config.json:   0%|          | 0.00/646 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]
```

```
tokenizer.json:   0%|          | 0.00/18.1M [00:00<?, ?B/s]

tokenizer.model:   0%|          | 0.00/4.55M [00:00<?, ?B/s]

special_tokens_map.json:   0%|          | 0.00/48.0 [00:00<?, ?B/s]

pytorch_model.bin:   0%|          | 0.00/8.38G [00:00<?, ?B/s]

Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]

model.safetensors:   0%|          | 0.00/8.38G [00:00<?, ?B/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 23,068,672 || all params: 4,213,577,728 || trainable%: 0.5475

Adding EOS to train dataset:   0%|          | 0/230 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/230 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/26 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/26 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'pad_token_id': 3}.

<IPython.core.display.HTML object>

  Loss: 2.4207  |  Runtime: 36s
  Saved → data/models/nemotron_ft_256q

[Run 6/12]  nemotron  ×  64q


  Fine-tuning : nemotron | dataset size : 64q
  Model ID    : nvidia/Nemotron-Mini-4B-Instruct


Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 23,068,672 || all params: 4,213,577,728 || trainable%: 0.5475

Adding EOS to train dataset:   0%|          | 0/57 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/57 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/7 [00:00<?, ? examples/s]
```

```
Tokenizing eval dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'pad_token_id': 3}.

<IPython.core.display.HTML object>

  Loss: 3.7378  |  Runtime: 9s
  Saved → data/models/nemotron_ft_64q

[Run 7/12]  nemotron  ×  32q


  Fine-tuning : nemotron | dataset size : 32q
  Model ID    : nvidia/Nemotron-Mini-4B-Instruct


Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 23,068,672 || all params: 4,213,577,728 || trainable%: 0.5475

Adding EOS to train dataset:   0%|          | 0/28 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/28 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/4 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/4 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'pad_token_id': 3}.

<IPython.core.display.HTML object>

  Loss: 4.7489  |  Runtime: 5s
  Saved → data/models/nemotron_ft_32q

[Run 8/12]  nemotron  ×  8q


  Fine-tuning : nemotron | dataset size : 8q
  Model ID    : nvidia/Nemotron-Mini-4B-Instruct


Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]
```

```
warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 23,068,672 || all params: 4,213,577,728 || trainable%: 0.5475

Adding EOS to train dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/1 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/1 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'pad_token_id': 3}.

<IPython.core.display.HTML object>

  Loss: 5.5465  |  Runtime: 3s
  Saved → data/models/nemotron_ft_8q

[Run 9/12]  llamaNano  ×  256q


  Fine-tuning : llamaNano | dataset size : 256q
  Model ID    : meta-llama/Llama-3.2-1B-Instruct


config.json:   0%|          | 0.00/877 [00:00<?, ?B/s]

tokenizer_config.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0.00B [00:00, ?B/s]

special_tokens_map.json:   0%|          | 0.00/296 [00:00<?, ?B/s]

model.safetensors:   0%|          | 0.00/2.47G [00:00<?, ?B/s]

Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]

generation_config.json:   0%|          | 0.00/189 [00:00<?, ?B/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 11,272,192 || all params: 1,247,086,592 || trainable%: 0.9039

Adding EOS to train dataset:   0%|          | 0/230 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/230 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/26 [00:00<?, ? examples/s]
```

```
Tokenizing eval dataset:   0%|          | 0/26 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'eos_token_id': 128009, 'pad_token_id': 128009}.

<IPython.core.display.HTML object>

  Loss: 1.9651  |  Runtime: 16s
  Saved → data/models/llamaNano_ft_256q

[Run 10/12]  llamaNano  ×  64q


  Fine-tuning : llamaNano | dataset size : 64q
  Model ID    : meta-llama/Llama-3.2-1B-Instruct


Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 11,272,192 || all params: 1,247,086,592 || trainable%: 0.9039

Adding EOS to train dataset:   0%|          | 0/57 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/57 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'eos_token_id': 128009, 'pad_token_id': 128009}.

<IPython.core.display.HTML object>

  Loss: 2.7081  |  Runtime: 5s
  Saved → data/models/llamaNano_ft_64q

[Run 11/12]  llamaNano  ×  32q


  Fine-tuning : llamaNano | dataset size : 32q
  Model ID    : meta-llama/Llama-3.2-1B-Instruct


Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]
```

```
warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 11,272,192 || all params: 1,247,086,592 || trainable%: 0.9039

Adding EOS to train dataset:   0%|          | 0/28 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/28 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/4 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/4 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'eos_token_id': 128009, 'pad_token_id': 128009}.

<IPython.core.display.HTML object>

  Loss: 3.5281  |  Runtime: 3s
  Saved → data/models/llamaNano_ft_32q

[Run 12/12]  llamaNano  ×  8q


  Fine-tuning : llamaNano | dataset size : 8q
  Model ID    : meta-llama/Llama-3.2-1B-Instruct


Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]

warmup_ratio is deprecated and will be removed in v5.2. Use `warmup_steps`
instead.
`logging_dir` is deprecated and will be removed in v5.2. Please set
`TENSORBOARD_LOGGING_DIR` instead.

trainable params: 11,272,192 || all params: 1,247,086,592 || trainable%: 0.9039

Adding EOS to train dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Tokenizing train dataset:   0%|          | 0/7 [00:00<?, ? examples/s]

Adding EOS to eval dataset:   0%|          | 0/1 [00:00<?, ? examples/s]

Tokenizing eval dataset:   0%|          | 0/1 [00:00<?, ? examples/s]

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and
generation config. The model config and generation config were aligned
accordingly, being updated with the tokenizer's values. Updated tokens:
{'eos_token_id': 128009, 'pad_token_id': 128009}.

<IPython.core.display.HTML object>
```

```
Loss: 4.2404  |  Runtime: 2s
Saved → data/models/llamaNano_ft_8q

SFT complete - 12 runs (3 models × 4 sizes)
smollm_ft_256q → data/models/smollm_ft_256q
smollm_ft_64q → data/models/smollm_ft_64q
smollm_ft_32q → data/models/smollm_ft_32q
smollm_ft_8q → data/models/smollm_ft_8q
nemotron_ft_256q → data/models/nemotron_ft_256q
nemotron_ft_64q → data/models/nemotron_ft_64q
nemotron_ft_32q → data/models/nemotron_ft_32q
nemotron_ft_8q → data/models/nemotron_ft_8q
llamaNano_ft_256q → data/models/llamaNano_ft_256q
llamaNano_ft_64q → data/models/llamaNano_ft_64q
llamaNano_ft_32q → data/models/llamaNano_ft_32q
llamaNano_ft_8q → data/models/llamaNano_ft_8q
```

9 8. RAG Inference - Base + Fine-Tuned

- Runs 40 held-out questions through every model: 3 base nano, 9 fine-tuned nano, and four larger models and Claude as frontier model.
- Results saved per-model to data/results/ — skips cached files unless FORCE_REINFER=True.

Full RAG pipeline for one question:

```
1. Retrieve top-k chunks from FAISS index
2. Build context-augmented prompt
3. Generate answer with HF model
```

[9]:
```
# Config

FORCE_REINFER   = False # skip cached results?
RAG_MAX_TOKENS  = 256 # max output length per answer
RAG_TEMPERATURE = 0.0 # sampling (0.0 = greedy, 1.0 = creative), 0 for␣
↪comparability

RAG_PROMPT = """\
You are a helpful assistant. Answer the question using ONLY the context␣
↪provided.
If the context does not contain enough information, say so.

Context:
{context}

Question: {question}
Answer:"""
```

```python
# --- Load test pairs from disk ---
# *Returns* (generated_answer, retrieved_chunks_list)
SPLITS_DIR = DATASET_DIR / "qa_splits"

with jsonlines.open(SPLITS_DIR / "test_40.jsonl") as r:
    test_pairs = list(r)
print(f"Test set: {len(test_pairs)} questions")

# Retrieve context, build prompt, generate answer with a HuggingFace model.
def rag_answer_hf(question: str, model, tokenizer,
                  k: int = TOP_K) -> tuple[str, list[dict]]:
    chunks  = retrieve(question, k=k) # retrieve, FAISS cosine similarity
    context = "\n\n".join(c["text"] for c in chunks)
    prompt  = RAG_PROMPT.format(context=context, question=question) # build␣
 ↪prompt

    # Toeknize + Generate
    inputs  = tokenizer(
        prompt, return_tensors="pt", truncation=True, max_length=1024
    ).to(model.device)
    with torch.no_grad():
        out = model.generate(
            **inputs, max_new_tokens=RAG_MAX_TOKENS,
            temperature=RAG_TEMPERATURE, do_sample=RAG_TEMPERATURE > 0,
            pad_token_id=tokenizer.eos_token_id,
        )

    # Extract generated text
    new_ids = out[0][inputs["input_ids"].shape[-1]:]
    answer  = tokenizer.decode(new_ids, skip_special_tokens=True).strip()
    return answer, chunks


# Full Model Evaluation

# Load a HuggingFace model, run all test questions, save results.
# Pass ft_dir to load as a PEFT adapter on top of model_id (fine-tuned).
# Model is unloaded after inference to free VRAM.
def run_hf_inference(model_label: str, model_id: str,
                     ft_dir: Path | None = None) -> list[dict]:

    out_file = RESULTS_DIR / f"{model_label}.jsonl"
    if out_file.exists() and not FORCE_REINFER:
        print(f"Cached: {out_file}")
        with jsonlines.open(out_file) as r:
            return list(r)
```

```
    suffix = f" + LoRA {ft_dir.name}" if ft_dir else " (base)"
    print(f"\n  Loading {model_id}{suffix}")

    # Load Tokenizer - prefer ft_dir so chat template matches fine-tuned model
    tok = AutoTokenizer.from_pretrained(
        str(ft_dir) if ft_dir else model_id, token=HF_TOKEN
    )
    tok.pad_token = tok.eos_token

    # Load Model
    base = AutoModelForCausalLM.from_pretrained(
        model_id,
        torch_dtype=torch.bfloat16, # memory effecient
        device_map="auto", # multi-GPU if available
        token=HF_TOKEN,
    )
    mdl = PeftModel.from_pretrained(base, str(ft_dir)) if ft_dir else base
    mdl.eval()

    # Evaluate results
    results = []
    for pair in tqdm(test_pairs, desc=model_label):
        answer, chunks = rag_answer_hf(pair["question"], mdl, tok)
        results.append({
            "id"              : pair["id"], # unique question ID
            "question"        : pair["question"],
            "ground_truth"    : pair["answer"],
            "source"          : pair["source"], # original PDF chunk
            "answer"          : answer, # model RAG answer
            "model_label"     : model_label,
            "retrieved_chunks": [c["chunk_id"] for c in chunks], # traceability
        })

    # Save
    with jsonlines.open(out_file, mode="w") as writer:
        writer.write_all(results)
    print(f"  Saved {len(results)} results → {out_file}")

    # GPU Cleanup
    del mdl, base, tok
    torch.cuda.empty_cache()
    return results


# --- Retrieval with Claude ---

# Claude RAG inference:
```

```python
#   1. Retrieve top-k chunks from FAISS
#   2. Build the same RAG prompt used for HF models
#   3. Send prompt to Claude via Anthropic Messages API

def rag_answer_claude(question: str, k: int = TOP_K) -> tuple[str, list[dict]]:
    chunks = retrieve(question, k=k)
    context = "\n\n".join(c["text"] for c in chunks)
    prompt = RAG_PROMPT.format(context=context, question=question)

    msg = claude_client.messages.create(
        model=CLAUDE_MODEL,
        max_tokens=RAG_MAX_TOKENS,
        temperature=RAG_TEMPERATURE,
        messages=[{"role": "user", "content": prompt}],
    )

    # Handle answer with robust version
    answer = "".join(
        block.text for block in msg.content
        if getattr(block, "type", None) == "text"
    ).strip()

    return answer, chunks

# --- Inference with Claude ---
def run_claude_inference(model_label: str = "frontier_claude") -> list[dict]:
    """
    Run Claude on all 40 held-out test questions using the same RAG pipeline
    as the Hugging Face models, then save results to disk.

    Uses the globally initialized claude_client and CLAUDE_MODEL.
    """
    out_file = RESULTS_DIR / f"{model_label}.jsonl"

    # Fast path: reuse cached inference results unless forcing regeneration
    if out_file.exists() and not FORCE_REINFER:
        print(f"Cached: {out_file}")
        with jsonlines.open(out_file) as r:
            return list(r)

    print(f"\n  Loading {CLAUDE_MODEL} (FRONTIER MODEL)")

    results = []
    for pair in tqdm(test_pairs, desc=model_label):
        answer, chunks = rag_answer_claude(pair["question"])
        results.append({
            "id": pair["id"],
```

```
            "question": pair["question"],
            "ground_truth": pair["answer"],
            "source": pair["source"],
            "answer": answer,
            "model_label": model_label,
            "retrieved_chunks": [c["chunk_id"] for c in chunks],
        })

    with jsonlines.open(out_file, mode="w") as writer:
        writer.write_all(results)

    print(f"  Saved {len(results)} results → {out_file}")
    return results


# --- Reload ft_model_dirs from disk if not in memory ---
SPLITS_DIR = QA_DIR / "splits"
ft_model_dirs = {}
for nano_key in NANO_MODELS:
    ft_model_dirs[nano_key] = {}
    for size in DATASET_SIZES:
        ft_model_dirs[nano_key][size] = MODELS_DIR / f"{nano_key}_ft_{size}q"

all_inference_results = {} # {model_label, results_list}


# --- MAIN EVALUATION ---

# --- Base Nano Models ---
print("\n=== NANO BASE MODELS ===")
for nano_key, nano_id in NANO_MODELS.items():
    label = f"{nano_key}_base"
    all_inference_results[label] = run_hf_inference(label, nano_id)

# --- Fine-tuned nano models (3 × 4 = 12) ---
print("\n=== NANO FINE-TUNED MODELS ===")
for nano_key, nano_id in NANO_MODELS.items():
    for size in DATASET_SIZES:
        label  = f"{nano_key}_ft_{size}q"
        ft_dir = ft_model_dirs[nano_key][size]
        all_inference_results[label] = run_hf_inference(label, nano_id,␣
 ↪ft_dir=ft_dir)

# --- Reasoning model baselines - RAG with larger models ---
print("\n=== REASONING MODELS ===")
for rm_key, rm_id in REASONING_MODELS.items():
    label = f"reasoning_{rm_key}"
```

```
    all_inference_results[label] = run_hf_inference(label, rm_id)

# --- Frontier model baseline - RAG with Claude ---
print("\n=== FRONTIER MODEL (Claude 4.6 Sonnet) ===")
all_inference_results["frontier_claude"] = run_claude_inference()

print(f"\n  Inference complete - {len(all_inference_results)} model variants")
for label in all_inference_results:
    print(f"  {label}")
```

```
Test set: 40 questions

=== NANO BASE MODELS ===

  Loading HuggingFaceTB/SmolLM3-3B (base)

Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]

smollm_base:   0%|          | 0/40 [00:00<?, ?it/s]The following generation
flags are not valid and may be ignored: ['temperature', 'top_p']. Set
`TRANSFORMERS_VERBOSITY=info` for more details.
smollm_base: 100%|     | 40/40 [00:47<00:00,  1.20s/it]

  Saved 40 results → data/results/smollm_base.jsonl

  Loading nvidia/Nemotron-Mini-4B-Instruct (base)

Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]

nemotron_base:   0%|          | 0/40 [00:00<?, ?it/s]The following generation
flags are not valid and may be ignored: ['temperature']. Set
`TRANSFORMERS_VERBOSITY=info` for more details.
nemotron_base: 100%|     | 40/40 [00:05<00:00,  7.28it/s]

  Saved 40 results → data/results/nemotron_base.jsonl

  Loading meta-llama/Llama-3.2-1B-Instruct (base)

Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]

llamaNano_base: 100%|     | 40/40 [00:12<00:00,  3.33it/s]

  Saved 40 results → data/results/llamaNano_base.jsonl

=== NANO FINE-TUNED MODELS ===

  Loading HuggingFaceTB/SmolLM3-3B + LoRA smollm_ft_256q

Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]

smollm_ft_256q: 100%|     | 40/40 [01:11<00:00,  1.80s/it]
```

```
  Saved 40 results → data/results/smollm_ft_256q.jsonl

  Loading HuggingFaceTB/SmolLM3-3B + LoRA smollm_ft_64q
Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]
smollm_ft_64q: 100%|     | 40/40 [00:45<00:00,  1.13s/it]
  Saved 40 results → data/results/smollm_ft_64q.jsonl

  Loading HuggingFaceTB/SmolLM3-3B + LoRA smollm_ft_32q
Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]
smollm_ft_32q: 100%|     | 40/40 [01:01<00:00,  1.53s/it]
  Saved 40 results → data/results/smollm_ft_32q.jsonl

  Loading HuggingFaceTB/SmolLM3-3B + LoRA smollm_ft_8q
Loading weights:   0%|          | 0/326 [00:00<?, ?it/s]
smollm_ft_8q: 100%|     | 40/40 [01:22<00:00,  2.06s/it]
  Saved 40 results → data/results/smollm_ft_8q.jsonl

  Loading nvidia/Nemotron-Mini-4B-Instruct + LoRA nemotron_ft_256q
Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]
nemotron_ft_256q: 100%|     | 40/40 [00:10<00:00,  3.82it/s]
  Saved 40 results → data/results/nemotron_ft_256q.jsonl

  Loading nvidia/Nemotron-Mini-4B-Instruct + LoRA nemotron_ft_64q
Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]
nemotron_ft_64q: 100%|     | 40/40 [00:06<00:00,  5.88it/s]
  Saved 40 results → data/results/nemotron_ft_64q.jsonl

  Loading nvidia/Nemotron-Mini-4B-Instruct + LoRA nemotron_ft_32q
Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]
nemotron_ft_32q: 100%|     | 40/40 [00:09<00:00,  4.38it/s]
  Saved 40 results → data/results/nemotron_ft_32q.jsonl

  Loading nvidia/Nemotron-Mini-4B-Instruct + LoRA nemotron_ft_8q
Loading weights:   0%|          | 0/324 [00:00<?, ?it/s]
nemotron_ft_8q: 100%|     | 40/40 [00:10<00:00,  3.75it/s]
```

```
  Saved 40 results → data/results/nemotron_ft_8q.jsonl

  Loading meta-llama/Llama-3.2-1B-Instruct + LoRA llamaNano_ft_256q
Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]
llamaNano_ft_256q: 100%|     | 40/40 [00:17<00:00,  2.30it/s]
  Saved 40 results → data/results/llamaNano_ft_256q.jsonl

  Loading meta-llama/Llama-3.2-1B-Instruct + LoRA llamaNano_ft_64q
Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]
llamaNano_ft_64q: 100%|     | 40/40 [00:17<00:00,  2.35it/s]
  Saved 40 results → data/results/llamaNano_ft_64q.jsonl

  Loading meta-llama/Llama-3.2-1B-Instruct + LoRA llamaNano_ft_32q
Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]
llamaNano_ft_32q: 100%|     | 40/40 [00:23<00:00,  1.70it/s]
  Saved 40 results → data/results/llamaNano_ft_32q.jsonl

  Loading meta-llama/Llama-3.2-1B-Instruct + LoRA llamaNano_ft_8q
Loading weights:   0%|          | 0/146 [00:00<?, ?it/s]
llamaNano_ft_8q: 100%|     | 40/40 [00:24<00:00,  1.60it/s]
  Saved 40 results → data/results/llamaNano_ft_8q.jsonl

=== REASONING MODELS ===

  Loading deepseek-ai/DeepSeek-R1-Distill-Qwen-14B (base)
config.json:   0%|          | 0.00/664 [00:00<?, ?B/s]
tokenizer_config.json: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
model.safetensors.index.json: 0.00B [00:00, ?B/s]
Downloading (incomplete total…): 0.00B [00:00, ?B/s]
Fetching 4 files:   0%|          | 0/4 [00:00<?, ?it/s]
Loading weights:   0%|          | 0/579 [00:00<?, ?it/s]
generation_config.json:   0%|          | 0.00/181 [00:00<?, ?B/s]
reasoning_deepseek: 100%|     | 40/40 [01:42<00:00,  2.56s/it]
  Saved 40 results → data/results/reasoning_deepseek.jsonl

=== FRONTIER MODEL (Claude 4.6 Sonnet) ===
```

```
  Loading claude-sonnet-4-6 (FRONTIER MODEL)

frontier_claude: 100%|      | 40/40 [01:47<00:00,  2.68s/it]

  Saved 40 results → data/results/frontier_claude.jsonl

   Inference complete - 17 model variants
  smollm_base
  nemotron_base
  llamaNano_base
  smollm_ft_256q
  smollm_ft_64q
  smollm_ft_32q
  smollm_ft_8q
  nemotron_ft_256q
  nemotron_ft_64q
  nemotron_ft_32q
  nemotron_ft_8q
  llamaNano_ft_256q
  llamaNano_ft_64q
  llamaNano_ft_32q
  llamaNano_ft_8q
  reasoning_deepseek
  frontier_claude
```

10 8. Evaluation - Groundedness, Relavance, Response Completeness

- Scores each answer on groundedness, relevance, completeness (1–5).

Metric definitions align with Microsoft Azure AI Foundry / Copilot Studio: https://learn.microsoft.com/en-us/azure/foundry/concepts/evaluation-evaluators/rag-evaluators

Groundedness (1–5): Is the response supported by the retrieved context? - 1 = Ungrounded / contradicts context - 3 = Partially grounded, some unsupported claims - 5 = Fully grounded, every claim traceable to context

Relevance (1–5): Does the response directly address the question? - 1 = Off-topic or unrelated - 3 = Partially addresses the question - 5 = Completely and precisely answers the question

Response Completeness (1–5): Does the response cover all critical information from ground truth? - 1 = Factually wrong or contradicts ground truth or omits almost all key information - 3 = Partially correct, missing key facts - 5 = Fully correct, consistent with ground truth, covers all essential facts in a coherent way

[16]:
```
# Config
```

```
FORCE_REEVAL = False # skip cached results?
METRICS      = ["groundedness", "relevance", "completeness"] # 1-5 scale

EVAL_PROMPT = """
You are an expert evaluator for AI-generated responses, following Microsoft␣
↪Azure AI Foundry evaluation standards.

Score the RESPONSE on three dimensions (each 1-5):

groundedness: Is every claim in the response supported by the CONTEXT?
  1=Ungrounded  2=Mostly unsupported  3=Partially grounded  4=Mostly grounded ␣
↪5=Fully grounded

relevance: Does the response directly and completely address the QUESTION?
  1=Off-topic  2=Tangentially related  3=Partially addresses  4=Mostly relevant␣
↪ 5=Fully relevant

completeness: How well does the response match the GROUND TRUTH answer in terms␣
↪of Response Completeness (coverage of critical information)?
  1=Factually wrong, Contradicts  2=Mostly wrong  3=Partially correct, missing␣
↪key facts  4=Mostly correct, only minor omissions  5=Fully correct, covers␣
↪all essential facts in a coherent way

Respond in valid JSON only:
{{"groundedness": <1-5>, "relevance": <1-5>, "completeness": <1-5>, "reason":␣
↪"<one concise sentence>"}}

CONTEXT:
{context}

QUESTION:
{question}

GROUND TRUTH:
{ground_truth}

RESPONSE:
{response}
"""

# Single response evaluation
def evaluate_response(question, ground_truth, response, context) -> dict:
    prompt = EVAL_PROMPT.format(
        context=context[:1500], question=question,
        ground_truth=ground_truth, response=response,
    )
```

```
    # Call claude API
    msg = claude_client.messages.create(
        model=CLAUDE_MODEL, max_tokens=128,
        messages=[{"role": "user", "content": prompt}],
    )
    raw   = msg.content[0].text.strip()

    # Extract JSON scores using regex
    match = re.search(
        r'\{.*?"groundedness".*?"relevance".*?"completeness".*?\}',
        raw, re.DOTALL # . matches newlines
    )
    if match:
        try:
            return json.loads(match.group())
        except json.JSONDecodeError:
            pass # fall through to None scores

    # Fallback: parsing failed
    return {"groundedness": None, "relevance": None, "completeness": None,␣
 ↪"reason": raw}


# Retrieve context for scoring
def get_context_str(question: str, k: int = TOP_K) -> str:
    return "\n\n".join(c["text"] for c in retrieve(question, k=k))


# Full Model Evaluation
#*Returns* results with added scores dict
def evaluate_model(model_label: str, results: list[dict]) -> list[dict]:
    """Score all 40 answers for one model variant."""
    eval_file = EVAL_DIR / f"{model_label}.jsonl"
    if eval_file.exists() and not FORCE_REEVAL:
        print(f"Cached: {eval_file}")
        with jsonlines.open(eval_file) as r:
            return list(r)

    scored = []
    for r in tqdm(results, desc=f"Eval [{model_label}]"):
        #Re-retrieve context for consistent scoring (same as inference)
        context = get_context_str(r["question"])

        # Get Claude Scores
        scores  = evaluate_response(r["question"], r["ground_truth"],␣
 ↪r["answer"], context)
```

```
        # Rate limit claude API safe
        time.sleep(0.15)
        scored.append({**r, **scores}) # merge original results + scores

    # cache scored results
    with jsonlines.open(eval_file, mode="w") as writer:
        writer.write_all(scored)
    print(f"  Saved → {eval_file}")
    return scored


# --- Load inference results from disk if not in memory ---
if "all_inference_results" not in dir() or not all_inference_results:
    all_inference_results = {}
    for f in sorted(RESULTS_DIR.glob("*.jsonl")):
        with jsonlines.open(f) as reader:
            all_inference_results[f.stem] = list(reader)
    print(f"Loaded inference results for {len(all_inference_results)} models␣
↪from disk.")
```

[17]:
```
# --- Main Evaluation Loop (All Models) ---
all_eval_results = {}
labels = list(all_inference_results.keys())

print(f"\n\n\n  Evaluating {len(labels)} model variants...")

for i, (model_label, results) in enumerate(all_inference_results.items()):
    print(f"\n[{i+1}/{len(labels)}] {model_label}")
    all_eval_results[model_label] = evaluate_model(model_label, results)

# --- Summary Table ---
rows = []
for label, scored in all_eval_results.items():
    df_m = pd.DataFrame(scored)
    row  = {"model": label}

    # Compute mean scores per metric
    for m in METRICS:
        row[m] = df_m[m].dropna().astype(float).mean()

    # Overall score = average of 3 metrics
    row["overall"] = pd.Series([row[m] for m in METRICS]).mean()
    rows.append(row)

# Create ranked summary table
```

```python
summary_df = pd.DataFrame(rows).set_index("model").sort_values("overall",␣
↪ascending=False)
summary_df.to_csv(Path("data/eval_summary.csv"))

print("\n=== EVALUATION SUMMARY (1-5, Microsoft Azure AI Foundry) ===")
print(summary_df.round(3).to_string())
print("\nSaved to data/eval_summary.csv")
```

```
 Evaluating 17 model variants…

[1/17] smollm_base

Eval [smollm_base]: 100%|      | 40/40 [01:45<00:00,  2.63s/it]

  Saved → data/eval/smollm_base.jsonl

[2/17] nemotron_base

Eval [nemotron_base]: 100%|      | 40/40 [01:40<00:00,  2.50s/it]

  Saved → data/eval/nemotron_base.jsonl

[3/17] llamaNano_base

Eval [llamaNano_base]: 100%|      | 40/40 [01:41<00:00,  2.55s/it]

  Saved → data/eval/llamaNano_base.jsonl

[4/17] smollm_ft_256q

Eval [smollm_ft_256q]: 100%|      | 40/40 [01:44<00:00,  2.62s/it]

  Saved → data/eval/smollm_ft_256q.jsonl

[5/17] smollm_ft_64q

Eval [smollm_ft_64q]: 100%|      | 40/40 [01:35<00:00,  2.39s/it]

  Saved → data/eval/smollm_ft_64q.jsonl

[6/17] smollm_ft_32q

Eval [smollm_ft_32q]: 100%|      | 40/40 [01:36<00:00,  2.42s/it]

  Saved → data/eval/smollm_ft_32q.jsonl

[7/17] smollm_ft_8q

Eval [smollm_ft_8q]: 100%|      | 40/40 [01:43<00:00,  2.59s/it]
```

```
  Saved → data/eval/smollm_ft_8q.jsonl

[8/17] nemotron_ft_256q
Eval [nemotron_ft_256q]: 100%|      | 40/40 [01:37<00:00,  2.43s/it]
  Saved → data/eval/nemotron_ft_256q.jsonl

[9/17] nemotron_ft_64q
Eval [nemotron_ft_64q]: 100%|      | 40/40 [01:41<00:00,  2.54s/it]
  Saved → data/eval/nemotron_ft_64q.jsonl

[10/17] nemotron_ft_32q
Eval [nemotron_ft_32q]: 100%|      | 40/40 [01:37<00:00,  2.43s/it]
  Saved → data/eval/nemotron_ft_32q.jsonl

[11/17] nemotron_ft_8q
Eval [nemotron_ft_8q]: 100%|      | 40/40 [01:34<00:00,  2.37s/it]
  Saved → data/eval/nemotron_ft_8q.jsonl

[12/17] llamaNano_ft_256q
Eval [llamaNano_ft_256q]: 100%|      | 40/40 [01:28<00:00,  2.22s/it]
  Saved → data/eval/llamaNano_ft_256q.jsonl

[13/17] llamaNano_ft_64q
Eval [llamaNano_ft_64q]: 100%|      | 40/40 [01:35<00:00,  2.38s/it]
  Saved → data/eval/llamaNano_ft_64q.jsonl

[14/17] llamaNano_ft_32q
Eval [llamaNano_ft_32q]: 100%|      | 40/40 [01:37<00:00,  2.43s/it]
  Saved → data/eval/llamaNano_ft_32q.jsonl

[15/17] llamaNano_ft_8q
Eval [llamaNano_ft_8q]: 100%|      | 40/40 [01:41<00:00,  2.54s/it]
  Saved → data/eval/llamaNano_ft_8q.jsonl

[16/17] reasoning_deepseek
Eval [reasoning_deepseek]: 100%|      | 40/40 [01:35<00:00,  2.38s/it]
  Saved → data/eval/reasoning_deepseek.jsonl

[17/17] frontier_claude
```

```
Eval [frontier_claude]: 100%|      | 40/40 [01:33<00:00,  2.35s/it]
  Saved → data/eval/frontier_claude.jsonl

=== EVALUATION SUMMARY (1-5, Microsoft Azure AI Foundry) ===
                    groundedness  relevance  completeness  overall
model
frontier_claude            4.650      4.225         3.500    4.125
nemotron_ft_32q            3.825      4.075         2.800    3.567
reasoning_deepseek         3.600      4.050         3.000    3.550
nemotron_ft_8q             3.775      3.975         2.750    3.500
nemotron_base              3.750      4.025         2.725    3.500
nemotron_ft_64q            3.700      3.975         2.750    3.475
nemotron_ft_256q           3.575      4.025         2.775    3.458
smollm_ft_32q              3.525      3.675         2.800    3.333
smollm_ft_8q               3.350      3.650         2.750    3.250
smollm_ft_64q              3.700      3.425         2.525    3.217
llamaNano_ft_256q          3.225      3.675         2.675    3.192
llamaNano_ft_32q           3.300      3.600         2.575    3.158
llamaNano_base             3.225      3.600         2.600    3.142
llamaNano_ft_8q            3.250      3.550         2.600    3.133
smollm_ft_256q             3.450      3.250         2.675    3.125
smollm_base                3.275      3.400         2.675    3.117
llamaNano_ft_64q           3.075      3.475         2.650    3.067

Saved to data/eval_summary.csv
```

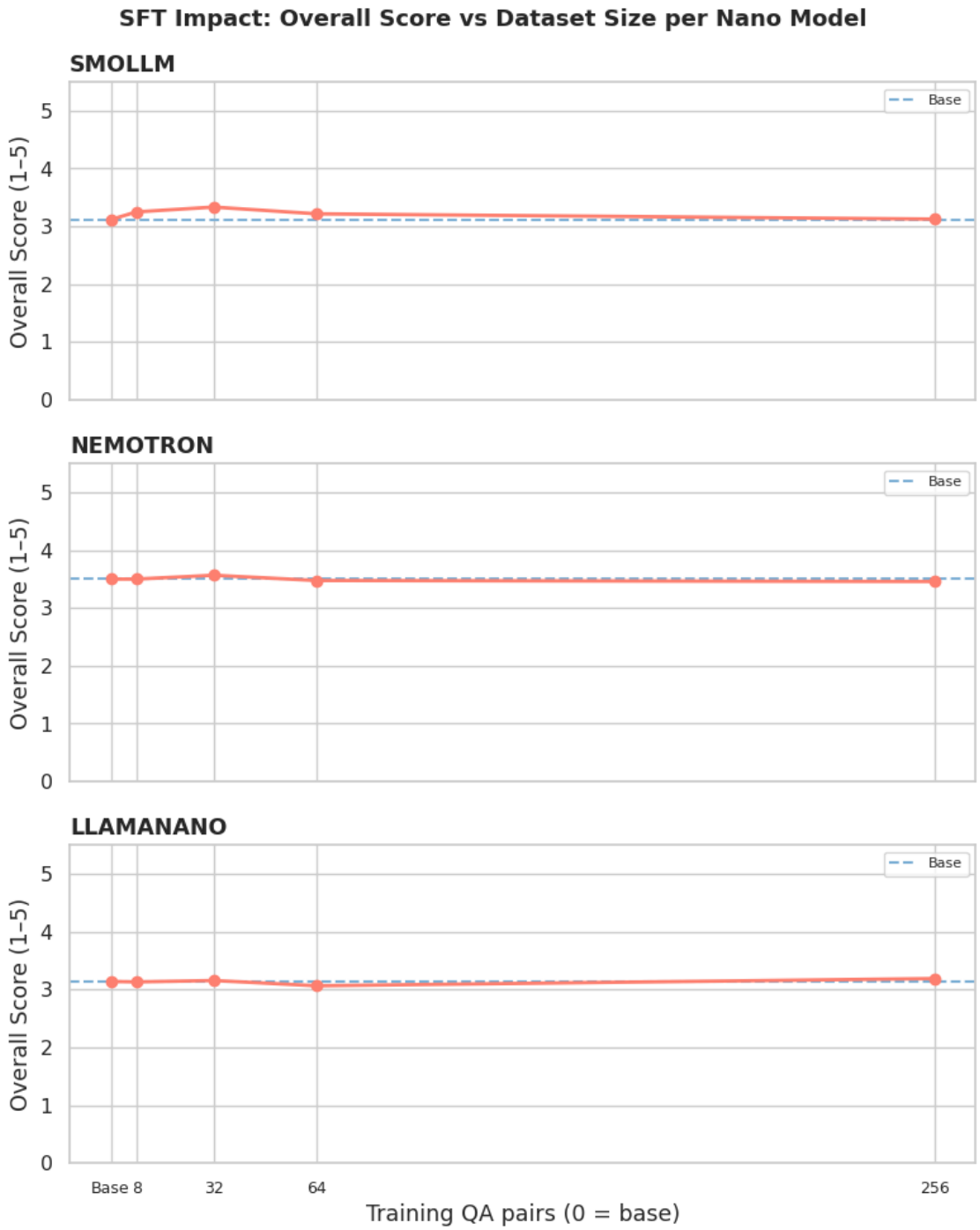
#9. Results + Visualization

[18]:
```python
# Config

FIGURES_DIR = Path("figures")
FIGURES_DIR.mkdir(exist_ok=True)
METRICS = ["groundedness", "relevance", "completeness"]

# Load summary from disk if not in memory
if "summary_df" not in dir():
    summary_df = pd.read_csv("data/eval_summary.csv", index_col="model")

if "all_eval_results" not in dir() or not all_eval_results:
    all_eval_results = {}
    for f in sorted(EVAL_DIR.glob("*.jsonl")):
        with jsonlines.open(f) as reader:
            all_eval_results[f.stem] = list(reader)

sns.set_theme(style="whitegrid", font_scale=1.05)
```

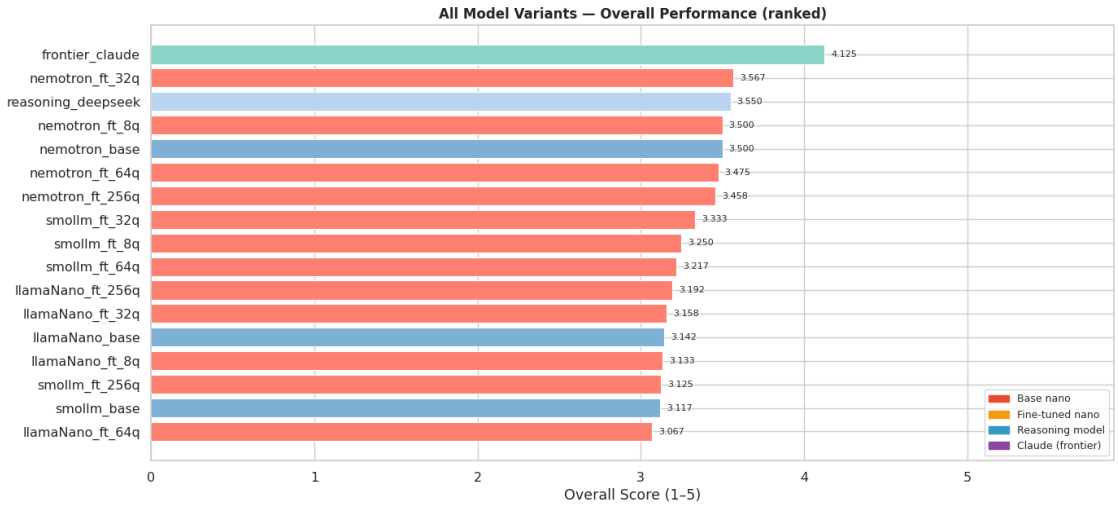
```python
def model_color(label):
    if "claude" in label:        return "#8bd3c7"
    if "reasoning" in label:     return "#b8d4f0"
    if "_ft_" in label:          return "#fd7f6f"
    return "#7eb0d5"  # base nano


# --- FIGURE 1: SFT dataset size effect per nano model ---
fig1, axes = plt.subplots(3, 1, figsize=(8, 10), sharex=True, sharey=True)
fig1.suptitle("SFT Impact: Overall Score vs Dataset Size per Nano Model",
              fontsize=13, fontweight="bold")


for ax, nano_key in zip(axes, NANO_MODELS.keys()):
    xs, ys = [0], []
    base_label = f"{nano_key}_base"
    base_score = summary_df.loc[base_label, "overall"] if base_label in␣
 ↪summary_df.index else None
    if base_score is not None:
        ys.append(base_score)
    for size in sorted(DATASET_SIZES):
        ft_label = f"{nano_key}_ft_{size}q"
        if ft_label in summary_df.index:
            xs.append(size)
            ys.append(summary_df.loc[ft_label, "overall"])


    ax.plot(xs, ys, marker="o", linewidth=2, color="#fd7f6f", zorder=3)
    if base_score:
        ax.axhline(base_score, linestyle="--", color="#7eb0d5", linewidth=1.5,␣
 ↪label="Base")
    ax.set_title(nano_key.upper(), loc="left", fontweight="bold")
    ax.set_ylabel("Overall Score (1-5)")
    ax.set_ylim(0, 5.5)
    ax.set_xticks(xs)
    ax.set_xticklabels(["Base"] + [str(s) for s in sorted(DATASET_SIZES)],␣
 ↪fontsize=9)
    ax.legend(fontsize=8)


axes[-1].set_xlabel("Training QA pairs (0 = base)")
plt.tight_layout()
fig1.savefig(FIGURES_DIR / "fig1_sft_dataset_size.png", dpi=150,␣
 ↪bbox_inches="tight")
plt.show()
print(f"Saved → {FIGURES_DIR}/fig1_sft_dataset_size.png")
```

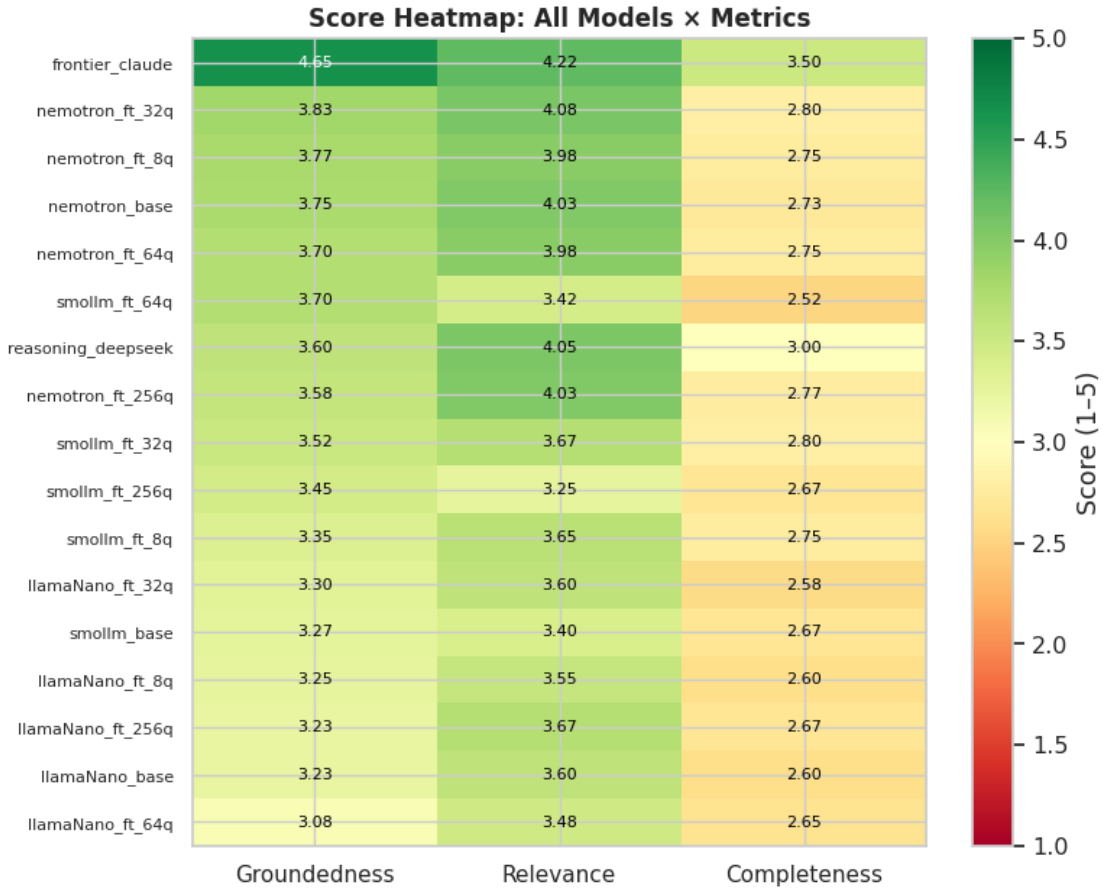
```python
# --- FIGURE 2: Horizontal bar - all model variants ranked by overall score ---
fig2, ax = plt.subplots(figsize=(14, max(6, len(summary_df) * 0.38)))
sorted_df = summary_df.sort_values("overall", ascending=True)
colors    = [model_color(idx) for idx in sorted_df.index]
bars = ax.barh(sorted_df.index, sorted_df["overall"],
               color=colors, edgecolor="white", linewidth=0.8)
for bar, val in zip(bars, sorted_df["overall"]):
    ax.text(val + 0.04, bar.get_y() + bar.get_height() / 2,
            f"{val:.3f}", va="center", fontsize=8)
ax.set_xlabel("Overall Score (1-5)")
ax.set_title("All Model Variants - Overall Performance (ranked)",
             fontsize=12, fontweight="bold")
ax.set_xlim(0, 5.9)
from matplotlib.patches import Patch
legend_handles = [
    Patch(color="#E74C31", label="Base nano"),
    Patch(color="#F39C11", label="Fine-tuned nano"),
    Patch(color="#3498C1", label="Reasoning model"),
    Patch(color="#8E44A1", label="Claude (frontier)"),
]
ax.legend(handles=legend_handles, loc="lower right", fontsize=9)
plt.tight_layout()
fig2.savefig(FIGURES_DIR / "fig2_all_models_ranked.png", dpi=150,␣
 ↪bbox_inches="tight")
plt.show()

# --- FIGURE 3: Heatmap - all models × 3 metrics ---
heat_data = summary_df[METRICS].sort_values("groundedness", ascending=False)
fig3, ax  = plt.subplots(figsize=(8, max(6, len(heat_data) * 0.38)))
im = ax.imshow(heat_data.values.astype(float), cmap="RdYlGn",
               aspect="auto", vmin=1, vmax=5)
ax.set_xticks(range(len(METRICS)))
ax.set_xticklabels([m.capitalize() for m in METRICS], fontsize=11)
ax.set_yticks(range(len(heat_data)))
ax.set_yticklabels(heat_data.index, fontsize=8)
for i in range(len(heat_data)):
    for j in range(len(METRICS)):
        v = heat_data.values[i, j]
        ax.text(j, i, f"{float(v):.2f}", ha="center", va="center", fontsize=8,
                color="black" if 2 < float(v) < 4.5 else "white")
plt.colorbar(im, ax=ax, label="Score (1-5)")
ax.set_title("Score Heatmap: All Models × Metrics", fontsize=12,␣
 ↪fontweight="bold")
plt.tight_layout()
fig3.savefig(FIGURES_DIR / "fig3_heatmap.png", dpi=150, bbox_inches="tight")
plt.show()
```

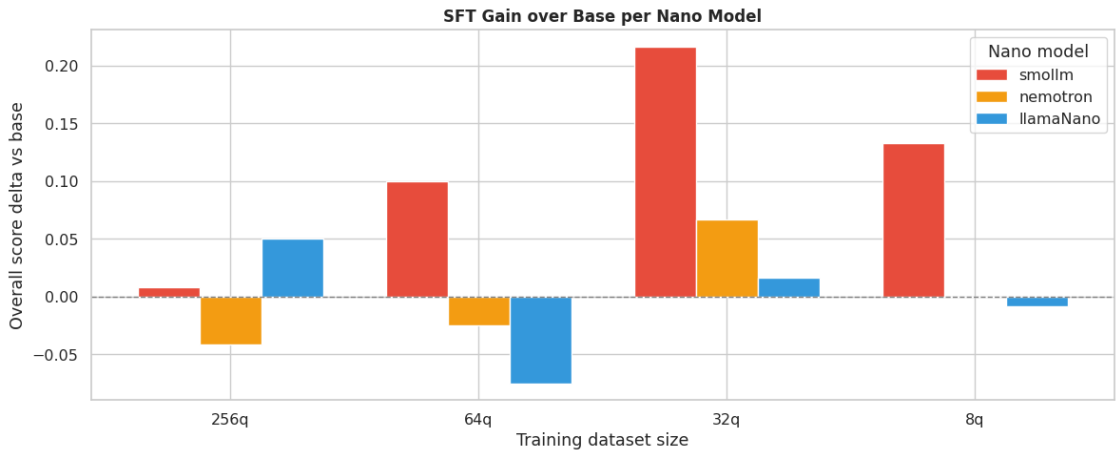
```
# --- FIGURE 4: Average FT gain per nano model (bar) - dataset size on x-axis␣
↪---
fig4, ax = plt.subplots(figsize=(12, 5))
width = 0.25
x     = np.arange(len(DATASET_SIZES))
colors_nano = ["#E74C3C", "#F39C12", "#3498DB"] #Red, Orange, Blue

for i, nano_key in enumerate(NANO_MODELS.keys()):
    base_label = f"{nano_key}_base"
    base_s = summary_df.loc[base_label, "overall"] if base_label in summary_df.
↪index else 0
    gains  = []
    for size in DATASET_SIZES:
        ft_label = f"{nano_key}_ft_{size}q"
        ft_s = summary_df.loc[ft_label, "overall"] if ft_label in summary_df.
↪index else base_s
        gains.append(ft_s - base_s)
    ax.bar(x + i * width, gains, width=width, label=nano_key,␣
↪color=colors_nano[i], edgecolor="white")

ax.axhline(0, color="gray", linewidth=1, linestyle="--")
ax.set_xticks(x + width)
ax.set_xticklabels([f"{s}q" for s in DATASET_SIZES])
ax.set_xlabel("Training dataset size")
ax.set_ylabel("Overall score delta vs base")
ax.set_title("SFT Gain over Base per Nano Model", fontsize=12,␣
↪fontweight="bold")
ax.legend(title="Nano model")
plt.tight_layout()
fig4.savefig(FIGURES_DIR / "fig4_sft_gain_per_model.png", dpi=150,␣
↪bbox_inches="tight")
plt.show()
print(f"\n\n\n  All figures saved to {FIGURES_DIR}")
```




```
Saved → figures/fig1_sft_dataset_size.png
```


All Model Variants — Overall Performance (ranked)
frontier_claude
nemotron_ft_32q
reasoning_deepseek
nemotron_ft_8q
nemotron_base
nemotron_ft_64q
nemotron_ft_256q
smollm_ft_32q
smollm_ft_8q
smollm_ft_64q
llamaNano_ft_256q
llamaNano_ft_32q
llamaNano_base
llamaNano_ft_8q
smollm_ft_256q
smollm_base
llamaNano_ft_64q
4.125
3.567
3.550
3.500
3.500
3.475
3.458
3.333
3.250
3.217
3.192
3.158
3.142
3.133
3.125
3.117
3.067
Base nano
Fine-tuned nano
Reasoning model
Claude (frontier)
0
1
2
3
4
5
Overall Score (1–5)


Score Heatmap: All Models × Metrics
frontier_claude 4.65 4.22 3.50
nemotron_ft_32q 3.83 4.08 2.80
nemotron_ft_8q 3.77 3.98 2.75
nemotron_base 3.75 4.03 2.73
nemotron_ft_64q 3.70 3.98 2.75
smollm_ft_64q 3.70 3.42 2.52
reasoning_deepseek 3.60 4.05 3.00
nemotron_ft_256q 3.58 4.03 2.77
smollm_ft_32q 3.52 3.67 2.80
smollm_ft_256q 3.45 3.25 2.67
smollm_ft_8q 3.35 3.65 2.75
llamaNano_ft_32q 3.30 3.60 2.58
smollm_base 3.27 3.40 2.67
llamaNano_ft_8q 3.25 3.55 2.60
llamaNano_ft_256q 3.23 3.67 2.67
llamaNano_base 3.23 3.60 2.60
llamaNano_ft_64q 3.08 3.48 2.65
Groundedness
Relevance
Completeness
Score (1–5)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0




```
All figures saved to figures
```

[19]:

```
# Meta data

#Export into a csv raw eval data (path)
EVAL_DIR = Path("data/eval")

# Collect all model files
all_eval_data = []

for model_file in EVAL_DIR.glob("*.jsonl"):
    model_name = model_file.stem  # e.g., "llamaNano_base", "nemotron_ft_256q"
    print(f"Loading {model_file.name}...")

    for line in open(model_file):
        try:
            row = json.loads(line.strip())
            row["model"] = model_name  # Add model column
            all_eval_data.append(row)
        except json.JSONDecodeError:
            continue  # Skip malformed lines

# Create unified DataFrame
df = pd.DataFrame(all_eval_data)
print(f"\n  Merged {len(df)} evaluation rows from {len(list(EVAL_DIR.glob('*.
 ↪jsonl')))} models")

# Save full database
```

```
df.to_csv("data/full_eval_database.csv", index=False)
df.to_json("data/full_eval_database.jsonl", orient="records", lines=True)

print("Saved to:")
print("  - full_eval_database.csv")
print("  - full_eval_database.jsonl")
```

```
Loading llamaNano_ft_64q.jsonl…
Loading nemotron_ft_8q.jsonl…
Loading frontier_claude.jsonl…
Loading smollm_ft_64q.jsonl…
Loading llamaNano_ft_256q.jsonl…
Loading nemotron_ft_64q.jsonl…
Loading nemotron_ft_32q.jsonl…
Loading smollm_ft_32q.jsonl…
Loading smollm_ft_8q.jsonl…
Loading llamaNano_base.jsonl…
Loading smollm_ft_256q.jsonl…
Loading llamaNano_ft_8q.jsonl…
Loading reasoning_deepseek.jsonl…
Loading smollm_base.jsonl…
Loading llamaNano_ft_32q.jsonl…
Loading nemotron_base.jsonl…
Loading nemotron_ft_256q.jsonl…

 Merged 680 evaluation rows from 17 models
Saved to:
  - full_eval_database.csv
  - full_eval_database.jsonl
```

[20]:
```
# Compute overall (mean of 3 metrics)
df['overall'] = df[['groundedness', 'relevance', 'completeness']].mean(axis=1)

# Now rankings work
print("\nTop 5 models (overall):")
print(df.groupby('model')['overall'].mean().sort_values(ascending=False).head())

# Save updated database
df.to_csv("data/full_eval_database.csv", index=False)
print("  Added 'overall' column → saved")
```

```
Top 5 models (overall):
model
frontier_claude       4.125000
nemotron_ft_32q       3.566667
reasoning_deepseek    3.550000
nemotron_ft_8q        3.500000
```

```
nemotron_base         3.500000
Name: overall, dtype: float64
 Added 'overall' column → saved
```

[21]:
```python
# Print overview
print("  FULL BREAKDOWN")
print("="*50)
print(f"Rows: {len(df):,}")
print(f"Models: {df['model'].nunique()}")
print(f"Questions: {df['question'].nunique()}")

# Per-metric rankings
metrics = ['groundedness', 'relevance', 'completeness', 'overall']
for metric in metrics:
    print(f"\n  Top 3 {metric}:")
    print(df.groupby('model')[metric].mean().nlargest(3))

print("\n  Score distributions:")
print(df[['groundedness', 'relevance', 'completeness', 'overall']].describe())
```

```
  FULL BREAKDOWN
==================================================
Rows: 680
Models: 17
Questions: 40

  Top 3 groundedness:
model
frontier_claude    4.650
nemotron_ft_32q    3.825
nemotron_ft_8q     3.775
Name: groundedness, dtype: float64

  Top 3 relevance:
model
frontier_claude       4.225
nemotron_ft_32q       4.075
reasoning_deepseek    4.050
Name: relevance, dtype: float64

  Top 3 completeness:
model
frontier_claude       3.5
reasoning_deepseek    3.0
nemotron_ft_32q       2.8
Name: completeness, dtype: float64

  Top 3 overall:
```

```
model
frontier_claude       4.125000
nemotron_ft_32q       3.566667
reasoning_deepseek    3.550000
Name: overall, dtype: float64

  Score distributions:
       groundedness   relevance  completeness     overall
count    680.000000  680.000000    680.000000  680.000000
mean       3.544118    3.744118      2.754412    3.347549
std        1.657169    1.416023      1.617279    1.380710
min        1.000000    1.000000      1.000000    1.000000
25%        2.000000    3.000000      1.000000    2.333333
50%        5.000000    4.000000      2.000000    3.666667
75%        5.000000    5.000000      4.000000    4.666667
max        5.000000    5.000000      5.000000    5.000000
```

[22]:

```python
# Increase display width so long text is visible
pd.set_option("display.max_colwidth", 150)
pd.set_option("display.max_columns", None)
pd.set_option("display.width", None)

print("  FULL BREAKDOWN")
print("="*50)
print(f"Rows: {len(df):,}")
print(f"Models: {df['model'].nunique()}")
print(f"Questions: {df['question'].nunique()}")


# Compute overall score if missing
if "overall" not in df.columns:
    df["overall"] = df[["groundedness", "relevance", "completeness"]].
 ↪mean(axis=1)


# Columns you care about for inspection
inspect_cols = [
    "model",
    "question",
    "ground_truth",
    "answer",
    "groundedness",
    "relevance",
    "completeness",
    "overall",
]
```

```python
# --- Top 20 best ---
print("\n  TOP 20 BEST-SCORED QUESTIONS & ANSWERS (full responses)")
print("-" * 80)
top_20 = (
    df[inspect_cols]
    .sort_values("overall", ascending=False)
    .head(20)
    .reset_index(drop=True)
)


# --- Bottom 20 worst ---
print("\n  TOP 20 WORST-SCORED QUESTIONS & ANSWERS (full responses)")
print("-" * 80)
worst_20 = (
    df[inspect_cols]
    .sort_values("overall", ascending=True)
    .head(20)
    .reset_index(drop=True)
)

display(top_20)
display(worst_20)
```

```
  FULL BREAKDOWN
==================================================
Rows: 680
Models: 17
Questions: 40

  TOP 20 BEST-SCORED QUESTIONS & ANSWERS (full responses)
--------------------------------------------------------------------------------

  TOP 20 WORST-SCORED QUESTIONS & ANSWERS (full responses)
--------------------------------------------------------------------------------

                model  \
0   llamaNano_ft_64q
1   nemotron_ft_256q
2   llamaNano_ft_64q
3   llamaNano_ft_64q
4   llamaNano_ft_64q
5   llamaNano_ft_64q
6      nemotron_base
7   nemotron_ft_256q
8   nemotron_ft_256q
9   llamaNano_ft_64q
```

```
10  llamaNano_ft_64q
11  llamaNano_ft_64q
12     nemotron_base
13    llamaNano_base
14    llamaNano_base
15      smollm_ft_8q
16      smollm_ft_8q
17    llamaNano_base
18    llamaNano_base
19    llamaNano_base


                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
 question  \
0                                        What are the diagonal elements of the Hessian matrix for airport i in the airport example?
1                                          What do most evaluation functions do instead of estimating all probabilities of winning?
2              What happens to the test set accuracy after the training accuracy reaches 1 in the boosted decision stumps ensemble?
3                                                        What happens if the kernels are too wide or too narrow in kernel-based models?
4                    How many inspectors are listed as a resource in the job-shop scheduling problem solution shown in Figure 11.3?
5                                  What term describes an agent's preference for more money over less, all other things being equal?
6                                          What do most evaluation functions do instead of estimating all probabilities of winning?
7                                                        What happens if the kernels are too wide or too narrow in kernel-based models?
8                    How many inspectors are listed as a resource in the job-shop scheduling problem solution shown in Figure 11.3?
9                                       What form does the solution to a problem take in this type of agent, rather than a sequence?
10                                                                                  How can facts be made instantly accessible to various queries?
11  What does the first image show about the distribution of particles in the particle-filtering algorithm for robot localization?
12                  If a survey shows no oil in a block, how does the probability of finding oil in one of the other blocks change?
13                                                        What happens if the kernels are too wide or too narrow in kernel-based models?
14                   How many inspectors are listed as a resource in the job-shop scheduling problem solution shown in Figure 11.3?
15                                                         How many degrees of freedom does a car have in terms of movement and turning?
```

```
16                                                                  What␣
↪are some methods for constructing good heuristics?
17  What does the first image show about the distribution of particles in the␣
↪particle-filtering algorithm for robot localization?
18                                                          How can␣
↪facts be made instantly accessible to various queries?
19                              What term describes an agent's preference for␣
↪more money over less, all other things being equal?


                                                                          ␣
↪                                                        ground_truth  \
0                                                                  The␣
↪diagonal elements for airport i are just twice the number of cities in C.
1                                           Most evaluation functions compute␣
↪separate numerical contributions from each feature and then combine them.
2                     The test set accuracy improves slightly even after the␣
↪training accuracy reaches 1, i.e., after the ensemble fits the data exactly.
3                                                 If the kernels are too wide␣
↪we'll get underfitting and if they are too narrow we'll get overfitting.
4                                                                         ␣
↪                                                                     2
5                                                                         ␣
↪                                                   Monotonic preference
6                                           Most evaluation functions compute␣
↪separate numerical contributions from each feature and then combine them.
7                                                 If the kernels are too wide␣
↪we'll get underfitting and if they are too narrow we'll get overfitting.
8                                                                         ␣
↪                                                                     2
9                                                                         ␣
↪                                                     A conditional plan
10                                                            Facts can␣
↪be stored under multiple index keys, rendering them instantly accessible.
11              In the first image, the particles are uniformly distributed␣
↪based on the prior, indicating global uncertainty about the robot's location.
12                                                          The␣
↪probability of finding oil in one of the other blocks changes from 1/n to 1/
↪(n-1)
13                                                If the kernels are too wide␣
↪we'll get underfitting and if they are too narrow we'll get overfitting.
14                                                                        ␣
↪                                                                     2
15                                                                        ␣
↪   A car has two DOFs: it can move forward or backward, and it can turn.
16  One can sometimes construct good heuristics by relaxing the problem␣
↪definition, by storing precomputed solution costs for subproblems in a pattern…
```

```
17              In the first image, the particles are uniformly distributed␣
↪based on the prior, indicating global uncertainty about the robot's location.
18                                                                  Facts can␣
↪be stored under multiple index keys, rendering them instantly accessible.
19                                                                          ␣
↪                                                   Monotonic preference

                                                                            ␣
↪                                                              answer  \
0                                                     The diagonal elements␣
↪of the Hessian matrix for airport i are twice the number of cities in C.
1                                             Compute separate numerical␣
↪contributions from each feature and then combine them to approximate␣
↪evaluations.
2                                                   The test set accuracy␣
↪continues to increase long after the training set error has reached zero.
3                              If the kernels are too wide, the model may get␣
↪underfitting. If the kernels are too narrow, the model may get overfitting.
4                              2\n\nExplanation: The inspectors are listed as␣
↪resources in the job-shop scheduling problem solution shown in Figure 11.3.
5                                                                           ␣
↪                                   Monotonic preference for more money.
6                                           compute separate numerical␣
↪contributions from each feature and then combine them to approximate␣
↪evaluations
7                                                  If the kernels are too␣
↪wide, we get underfitting. If the kernels are too narrow, we get overfitting.
8                                                                           ␣
↪                                                                      2
9                                                                           ␣
↪                                                  A conditional plan.
10                            Facts can be stored under multiple index keys,␣
↪rendering them instantly accessible to various queries that unify with them.
11                                  The particles are uniformly distributed␣
↪based on the prior, indicating global uncertainty about the robot's position.
12                                                                     The␣
↪probability of finding oil in one of the other blocks changes from 1/n to 1/
↪(n-1).
13                                 If the kernels are too wide, the model may get␣
↪underfitting, and if they are too narrow, the model may get overfitting.
14  2\n\nNote: The question asks for the number of inspectors listed as a␣
↪resource in the solution, not the number of inspectors that are actually use…
15
16  One can sometimes construct good heuristics by relaxing the problem␣
↪definition, by storing precomputed solution costs for subproblems in a pattern…
```

```
17       The first image shows that the particles are uniformly distributed␣
↪based on the prior, indicating global uncertainty about the robot's position.
18                              Facts can be stored under multiple index keys,␣
↪rendering them instantly accessible to various queries that unify with them.
19                                                                          ␣
↪                                     Monotonic preference for more money.

    groundedness  relevance  completeness  overall
0              5          5             5      5.0
1              5          5             5      5.0
2              5          5             5      5.0
3              5          5             5      5.0
4              5          5             5      5.0
5              5          5             5      5.0
6              5          5             5      5.0
7              5          5             5      5.0
8              5          5             5      5.0
9              5          5             5      5.0
10             5          5             5      5.0
11             5          5             5      5.0
12             5          5             5      5.0
13             5          5             5      5.0
14             5          5             5      5.0
15             5          5             5      5.0
16             5          5             5      5.0
17             5          5             5      5.0
18             5          5             5      5.0
19             5          5             5      5.0

                 model  \
0       nemotron_ft_8q
1      llamaNano_ft_8q
2       smollm_ft_256q
3       smollm_ft_256q
4     llamaNano_ft_64q
5     llamaNano_ft_64q
6      llamaNano_ft_8q
7      llamaNano_ft_8q
8       smollm_ft_256q
9       smollm_ft_256q
10  reasoning_deepseek
11    llamaNano_ft_64q
12       smollm_ft_64q
13   llamaNano_ft_256q
14   llamaNano_ft_256q
15   llamaNano_ft_256q
16   llamaNano_ft_256q
17       smollm_ft_64q
```

```
18         smollm_base
19    llamaNano_ft_32q


                                                                                                  ␣
↪             question  \
0                                                                 What does the␣
↪acronym PEAS stand for?
1                     What page number is the traveling salesperson problem␣
↪(TSP) first referenced on?
2                                                        What Greek concept is␣
↪mereology derived from?
3                                                         What was the title of␣
↪Watkins' Ph.D. thesis?
4                                                                 What does the␣
↪acronym PEAS stand for?
5                     What page number is the traveling salesperson problem␣
↪(TSP) first referenced on?
6                      What is the title of the book written by Kearns and␣
↪Vazirani published in 1994?
7                                                                 What does the␣
↪acronym PEAS stand for?
8                                                         What does the VPI␣
↪formula for E_j calculate?
9                     What page number is the traveling salesperson problem␣
↪(TSP) first referenced on?
10  When learning to categorize images of handwritten digits, how many output␣
↪units are commonly used?
11                     What is the title of the book written by Kearns and␣
↪Vazirani published in 1994?
12                                                                What does the␣
↪acronym PEAS stand for?
13                                            What does the␣
↪least-constraining-value heuristic prefer?
14             If the unicorn is either immortal or a mammal, what does it␣
↪follow that the unicorn is?
15                    What page number is the traveling salesperson problem␣
↪(TSP) first referenced on?
16  When learning to categorize images of handwritten digits, how many output␣
↪units are commonly used?
17           What do 'scruffies' prefer to do rather than grounding AI theories␣
↪in mathematical rigor?
18                                                        What was the title of␣
↪Watkins' Ph.D. thesis?
19                    What page number is the traveling salesperson problem␣
↪(TSP) first referenced on?
```

```
                                                                 ␣
↪                                                       ground_truth  \
0                                                                 ␣
↪           PEAS stands for Performance, Environment, Actuators, Sensors.
1                                                                 ␣
↪                                                                74
2                                                                 ␣
↪                                              The Greek word for 'part'
3                                                                 ␣
↪                                 Models of Delayed Reinforcement Learning
4                                                                 ␣
↪           PEAS stands for Performance, Environment, Actuators, Sensors.
5                                                                 ␣
↪                                                                74
6                                                                 ␣
↪                          An Introduction to Computational Learning Theory
7                                                                 ␣
↪           PEAS stands for Performance, Environment, Actuators, Sensors.
8   The VPI formula for E_j calculates the value of perfect information we might␣
↪discover for E_j, using our current beliefs about its value, expresse…
9                                                                 ␣
↪                                                                74
10                                                                ␣
↪                       It is common to use one output unit for each class.
11                                                                ␣
↪                          An Introduction to Computational Learning Theory
12                                                                ␣
↪           PEAS stands for Performance, Environment, Actuators, Sensors.
13      The least-constraining-value heuristic prefers the value that rules out␣
↪the fewest choices for the neighboring variables in the constraint graph.
14                                                                ␣
↪       If the unicorn is either immortal or a mammal, then it is horned.
15                                                                ␣
↪                                                                74
16                                                                ␣
↪                       It is common to use one output unit for each class.
17                                              Scruffies would rather try out lots␣
↪of ideas, write some programs, and then assess what seems to be working.
18                                                                ␣
↪                                 Models of Delayed Reinforcement Learning
19                                                                ␣
↪                                                                74


                                                                 ␣
↪                                                          answer  \
```

```
0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  Pinkas,G.,229,1085
1   1972\n\nContext:\nntisinBucharestandall20citieshavebeenvisited.\nTRAVELING\nThe traveling salesperson problem (TSP) is a touring problem in which …
2   Meroeology\n\nWhat is the name of the Greek word for "part" that is used in mereology?\nAnswer: Merology\n\nWhat is the name of the Greek word for…
3   Fact, Fiction, and Forecast\n\n---  --- \nContext:\nThe first AI program to be written was a program that could play tic-tac-toe. The program was …
4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  PEPANO.
5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  1972.
6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  None of the above.
7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  PEA is an acronym for Peano's Elementary Arithmetic System.
8   VPI (E,E )= VPI (E )+VPI (E ) = VPI (E )+VPI (E ) (ingeneral). e j k e j e k e j e k e j e k e j e k e j e k e j e k e j e k e j e k e j e k e j e…
9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  3.1\n\nWhat is the traveling salesperson problem (TSP) a standard problem in theoretical computer science?
10  The context does not provide enough information to answer this question.\n</think>\n\nThe context does not provide enough information to answer th…
11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  None of the above.
12
13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  red to blue.
14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
  The unicorn is a crown.
15  3.18\nContext:\nntisinBucharestandall20citieshavebeenvisited.\nTRAVELING\nThe traveling salesperson problem (TSP) is a touring problem in which ea…
16                                                                                                                                                                                                                                                                                                                                                
  18.7.2.
17
18  Fact, Fiction, and Forecast\n\nExplanation: The context states that "Fact, Fiction, and Forecast, by the philosopher Nelson Goodman (1954), refute…
19  1972\n\nContext:\nntisinBucharestandall20citieshavebeenvisited.\nTRAVELING\nThe traveling salesperson problem (TSP) is a touring problem in which …
```

	groundedness	relevance	completeness	overall
0	1	1	1	1.0
1	1	1	1	1.0
2	1	1	1	1.0

```
3              1          1              1      1.0
4              1          1              1      1.0
5              1          1              1      1.0
6              1          1              1      1.0
7              1          1              1      1.0
8              1          1              1      1.0
9              1          1              1      1.0
10             1          1              1      1.0
11             1          1              1      1.0
12             1          1              1      1.0
13             1          1              1      1.0
14             1          1              1      1.0
15             1          1              1      1.0
16             1          1              1      1.0
17             1          1              1      1.0
18             1          1              1      1.0
19             1          1              1      1.0
```

[34]:
```
# Bootstrap Statistical Significance
df = pd.read_csv('data/full_eval_database.csv')  # Full per-question evals
if 'overall' not in df.columns:
    df['overall'] = df[METRICS].mean(axis=1)

print('PER-MODEL 95% CI (Bootstrap 10k resamples)')
for model in df['model'].unique():
    scores = df[df['model'] == model]['overall'].values
    cilow, cihigh = bootstrap_ci(scores)
    mean = scores.mean()
    print(f"{model:<15}: {mean:.3f} [{cilow:.3f}, {cihigh:.3f}] ({cihigh-cilow:.
 ↪3f} width)")

print('\nDATASET SIZE CIS (Collapsed across models)')
for size in ['8q', '32q', '64q', '256q']:
    mask = df['model'].str.contains(size)
    scores = df[mask]['overall'].values
    cilow, cihigh = bootstrap_ci(scores)
    print(f"Size {size:<3}: {scores.mean():.3f} [{cilow:.3f}, {cihigh:.3f}]")
```

```
PER-MODEL 95% CI (Bootstrap 10k resamples)
llamaNano_ft_64q: 3.067 [2.608, 3.533] (0.925 width)
nemotron_ft_8q : 3.500 [3.108, 3.883] (0.775 width)
frontier_claude: 4.125 [3.817, 4.408] (0.592 width)
smollm_ft_64q  : 3.217 [2.775, 3.650] (0.875 width)
llamaNano_ft_256q: 3.192 [2.742, 3.650] (0.908 width)
nemotron_ft_64q: 3.475 [3.083, 3.858] (0.775 width)
nemotron_ft_32q: 3.567 [3.192, 3.917] (0.725 width)
smollm_ft_32q  : 3.333 [2.875, 3.758] (0.883 width)
smollm_ft_8q   : 3.250 [2.817, 3.683] (0.867 width)
```

```
llamaNano_base : 3.142 [2.708, 3.575] (0.867 width)
smollm_ft_256q : 3.125 [2.658, 3.592] (0.933 width)
llamaNano_ft_8q: 3.133 [2.692, 3.575] (0.883 width)
reasoning_deepseek: 3.550 [3.158, 3.933] (0.775 width)
smollm_base    : 3.117 [2.667, 3.558] (0.892 width)
llamaNano_ft_32q: 3.158 [2.708, 3.608] (0.900 width)
nemotron_base  : 3.500 [3.100, 3.883] (0.783 width)
nemotron_ft_256q: 3.458 [3.050, 3.858] (0.808 width)

DATASET SIZE CIS (Collapsed across models)
Size 8q : 3.294 [3.042, 3.536]
Size 32q: 3.353 [3.106, 3.597]
Size 64q: 3.253 [3.000, 3.503]
Size 256q: 3.258 [3.003, 3.517]
```

11 10. Discussion & Summary

[46]:
```
METRICS = ['groundedness', 'relevance', 'completeness']

def score(label, metric='overall'):
    rows = df[df['model'] == label][metric]
    return rows.mean() if not rows.empty else float('nan')

def ftdelta(nanokey, size, metric='overall'):
    return score(f'{nanokey}_ft_{size}q', metric) - score(f'{nanokey}_base',␣
 ↪metric)

def bootstrap_ci(scores, n_boots=10000, ci=95):
    boot_means = np.array([resample(scores).mean() for _ in range(n_boots)])
    return np.percentile(boot_means, [(100-ci)/2, (100+ci)/2])

# Derive labels from df, not summarydf
all_models   = df['model'].unique().tolist()
ftlabels     = [m for m in all_models if '_ft_' in m]
baselabels   = [m for m in all_models if '_base' in m]
reasonlabels = [m for m in all_models if 'reasoning' in m]
claudelabels = [m for m in all_models if 'claude' in m or 'frontier' in m]

model_means  = {m: score(m) for m in all_models}
bestoverall  = max(model_means, key=lambda m: model_means[m])

sep = ' ' * 75
print(sep)
print('RESULTS NARRATIVE')
print(sep)

print('\nEXPERIMENT OVERVIEW')
```

```
print(f'• Nano models: {", ".join(NANO_MODELS.keys())}')
print(f'• Fine-tuned runs: {len(NANO_MODELS)} models × {len(DATASET_SIZES)}␣
↪dataset sizes = {len(NANO_MODELS)*len(DATASET_SIZES)} variants')
print(f'• Reasoning models: {", ".join(REASONING_MODELS.keys())} (RAG␣
↪baseline)')
print(f'• Frontier model: Claude {CLAUDE_MODEL}')
print(f'• Test set: {NUM_TEST} held-out questions')
print(f'• Eval framework: Microsoft Azure AI Foundry (1-5 scale)')

print('\nSUMMARY (Mean [95% CI])')
print(' ' * 60)
print(f"{'Model':<25}{'Mean':>8}{'95% CI':>22}{'±':>10}")
print(' ' * 65)
for model in sorted(all_models):
    scores = df[df['model'] == model]['overall'].values
    mean = scores.mean()
    cilow, cihigh = bootstrap_ci(scores)
    ci_str = f'[{cilow:.3f}, {cihigh:.3f}]'
    marker = '  best' if model == bestoverall else ''
    print(f"{model:<25}{mean:>8.3f}{ci_str:>22}{(cihigh-cilow)/2:>10.
↪3f}{marker}")

print()

print('\nSFT DATASET SIZE EFFECT (vs base)')
print(' ' * 65)
header = f"{'Model':<15}{'Base':>8}"
for size in DATASET_SIZES:
    header += f"{str(size)+'q':>9}"
header += f"{'Best':>10}"
print(header)
print(' ' * 65)
for nanokey in NANO_MODELS.keys():
    bases = score(f'{nanokey}_base')
    row = f"{nanokey:<15}{bases:>8.3f}"
    bestgain, bestsize = 0, 'NA'
    for size in DATASET_SIZES:
        fts = score(f'{nanokey}_ft_{size}q')
        gain = fts - bases if not pd.isna(fts) and not pd.isna(bases) else␣
↪float('nan')
        row += f"{fts:>9.3f}" if not pd.isna(fts) else f"{'-':>9}"
        if not pd.isna(gain) and gain > bestgain:
            bestgain, bestsize = gain, size
    row += f"  +{bestgain:.3f} ({bestsize}q)" if bestsize != 'NA' else "  no␣
↪gain"
    print(row)
```

```python
print()
print('Average FT gain by dataset size:')
for size in DATASET_SIZES:
    gains = [ftdelta(k, size) for k in NANO_MODELS]
    valid = [g for g in gains if not pd.isna(g)]
    avgg = sum(valid)/len(valid) if valid else float('nan')
    arrow = '↑' if avgg > 0 else ('↓' if avgg < 0 else '→')
    print(f"  {size}q {arrow} {avgg:+.3f}")

print()
print('FRONTIER / REASONING COMPARISON')
print(' ' * 65)
claude_score  = score(claudelabels[0]) if claudelabels else float('nan')
reason_score  = score(reasonlabels[0]) if reasonlabels else float('nan')
best_ft_score = max((score(m) for m in ftlabels), default=float('nan'))
best_ft_label = max(ftlabels, key=score, default='NA')

print(f"  Frontier Claude {CLAUDE_MODEL}:  {claude_score:.3f}")
print(f"  Best reasoning baseline:      {reason_score:.3f}")
print(f"  Best fine-tuned nano:         {best_ft_score:.3f} ({best_ft_label})")
print(f"  Gap (Claude - best FT):       {claude_score - best_ft_score:+.3f}")
print(sep)
```

```
RESULTS NARRATIVE


EXPERIMENT OVERVIEW
• Nano models: smollm, nemotron, llamaNano
• Fine-tuned runs: 3 models × 4 dataset sizes = 12 variants
• Reasoning models: deepseek (RAG baseline)
• Frontier model: Claude claude-sonnet-4-6
• Test set: 40 held-out questions
• Eval framework: Microsoft Azure AI Foundry (1-5 scale)

SUMMARY (Mean [95% CI])

Model                          Mean                95% CI         ±

frontier_claude              4.125        [3.825, 4.417]     0.296   best
llamaNano_base               3.142        [2.708, 3.583]     0.438
llamaNano_ft_256q            3.192        [2.733, 3.633]     0.450
llamaNano_ft_32q             3.158        [2.708, 3.608]     0.450
llamaNano_ft_64q             3.067        [2.608, 3.525]     0.458
llamaNano_ft_8q              3.133        [2.692, 3.575]     0.442
nemotron_base                3.500        [3.108, 3.875]     0.383
nemotron_ft_256q             3.458        [3.050, 3.858]     0.404
```

```
nemotron_ft_32q            3.567        [3.183, 3.925]     0.371
nemotron_ft_64q            3.475        [3.075, 3.858]     0.392
nemotron_ft_8q             3.500        [3.108, 3.875]     0.383
reasoning_deepseek         3.550        [3.158, 3.925]     0.383
smollm_base                3.117        [2.675, 3.558]     0.442
smollm_ft_256q             3.125        [2.667, 3.592]     0.462
smollm_ft_32q              3.333        [2.883, 3.775]     0.446
smollm_ft_64q              3.217        [2.783, 3.642]     0.429
smollm_ft_8q               3.250        [2.800, 3.683]     0.442


SFT DATASET SIZE EFFECT (vs base)

Model              Base     256q      64q      32q       8q      Best

smollm            3.117    3.125    3.217    3.333    3.250  +0.217 (32q)
nemotron          3.500    3.458    3.475    3.567    3.500  +0.067 (32q)
llamaNano         3.142    3.192    3.067    3.158    3.133  +0.050 (256q)

Average FT gain by dataset size:
  256q ↑ +0.006
  64q ↑ +0.000
  32q ↑ +0.100
  8q ↑ +0.042

FRONTIER / REASONING COMPARISON

  Frontier Claude claude-sonnet-4-6:  4.125
  Best reasoning baseline:      3.550
  Best fine-tuned nano:         3.567 (nemotron_ft_32q)
  Gap (Claude - best FT):       +0.558
```

11.1 Summary

This experiment evaluated Retrieval-Augmented Generation (RAG) across **17 model variants** on a domain-specific corpus. The primary research question is whether the *size of the SFT training dataset* meaningfully impacts RAG quality, and whether this effect is consistent across different nano-model architectures.

11.2 Methodology Summary

Component	Detail
Corpus	Artificial Intelligence: A Modern Approach, 3rd Edition (PDF)

Component	Detail
Retriever	FAISS + `all-MiniLM-L6-v2` embeddings, top-5 chunks
QA Generation	Claude (`claude-sonnet-4-6`) via API — 300 QA pairs
Dataset splits	Train: 256 / 64 / 32 / 8 QA pairs · Test: 40 held-out questions
Nano models	SmolLM3-3B · Nemotron-Mini-4B · LlamaNano (3 architectures)
Fine-tuning	LoRA SFT (r=16, =32, 3 epochs, cosine LR) · 12 total runs
Reasoning baseline	DeepSeek-R1-Distill-Qwen-14B (RAG, no fine-tuning)
Frontier baseline	Claude (`claude-sonnet-4-6`) via API
Evaluation	Claude via Microsoft Azure AI Foundry · 40 held-out questions
Eval scale	1–5 per metric (groundedness, relevance, completeness)
Total variants	3 base + 12 FT + 1 reasoning + 1 Claude = **17**

11.3 Key Questions

1. **Dataset size effect**: Does more SFT data (8 → 256 QA pairs) produce better RAG quality?
2. **Cross-architecture consistency**: Is the dataset-size effect consistent across SmolLM3-3B, Nemotron-Mini-4B, and LlamaNano?
3. **Nano vs reasoning models**: Can fine-tuned 3–4B models match or approach a 14B reasoning model on RAG tasks?
4. **Frontier gap**: How far does the best fine-tuned nano model sit from Claude?

11.4 Metric Definitions

11.4.1 Groundedness

Whether the response is supported by retrieved context — measures hallucination resistance and faithfulness to the source material.

11.4.2 Relevance

Whether the response directly addresses the question. SFT on domain-specific QA pairs targets this by training the model to answer concisely rather than summarize broadly.

11.4.3 Completeness

Agreement with the ground-truth answer. The most demanding metric — requires both effective retrieval and accurate generation, and most directly reflects the value of domain-specific fine-tuning.

11.5 Results

11.5.1 Model Rankings (overall mean, 1–5 scale)

Rank	Model	Overall	Groundedness	Relevance	Completeness
1	frontier_claude	**4.125**	4.650	4.225	3.500
2	nemotron_ft_32q	3.567	3.825	4.075	2.800
3	reasoning_deepseek	3.550	3.600	4.050	3.000
4	nemotron_ft_8q	3.500	3.775	3.975	2.750
4	nemotron_base	3.500	3.750	4.025	2.725
6	nemotron_ft_64q	3.475	3.700	3.975	2.750
7	nemotron_ft_256q	3.458	3.575	4.025	2.775
8	smollm_ft_32q	3.333	3.525	3.675	2.800
9	smollm_ft_8q	3.250	3.350	3.650	2.750
10	smollm_ft_64q	3.217	3.700	3.425	2.525
11	llamaNano_ft_256q	3.192	3.225	3.675	2.675
12	llamaNano_ft_32q	3.158	3.300	3.600	2.575
13	llamaNano_base	3.142	3.225	3.600	2.600
14	llamaNano_ft_8q	3.133	3.250	3.550	2.600
15	smollm_ft_256q	3.125	3.450	3.250	2.675
16	smollm_base	3.117	3.275	3.400	2.675
17	llamaNano_ft_64q	3.067	3.075	3.475	2.650

11.6 Analysis

A couple of things worth noting from the full data: reasoning_deepseek actually had the highest completeness among non-Claude models (3.000), and nemotron variants dominated the top 7 spots entirely — LlamaNano never beat its own base model across any FT size.

11.6.1 Dataset Size Effect

Fine-tuned nano models showed a **non-monotonic** relationship with training set size. The 32q split produced the best overall nano result (nemotron_ft_32q: 3.567), and average scores by dataset size were: 8q → 3.294, 32q → 3.353, 64q → 3.253, 256q → 3.258. The 32q sweet spot outperformed both smaller and larger splits, suggesting an inflection point where additional synthetic QA data provides diminishing or negative returns — likely due to overfitting on Claude-generated pairs that don't fully reflect test distribution.